



# KTU NOTES

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## MODULE III

### COMPLEX VARIABLE - DIFFERENTIATION

(Text 2: Sections 13.3, 13.4, 17.1, 17.2, 17.4)

Syllabus: Complex function, Limit, Continuity, derivative, analytic functions, Cauchy-Riemann equations, harmonic functions, finding harmonic conjugate, Conformal mappings — Mappings  $w = z^2$ ,  $w = e^z$ , Linear fractional transformation  $w = \frac{1}{z}$ , fixed points, Transformation  $w = \sin z$ .

### INTRODUCTION:

Complex Number: A complex number 'z' is an ordered pair  $(x, y)$  of real numbers  $x$  and  $y$ , written

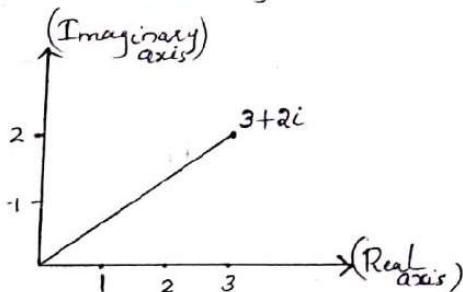
$$z = (x, y) = x + iy \text{ where } i = \sqrt{-1}$$

$x$  is called the real part and  $y$  is called the imaginary part of  $z$ , written

$$x = \operatorname{Re}(z), \quad y = \operatorname{Im}(z)$$

Complex plane: The  $xy$ -plane in which a given complex number  $z = x + iy$  represented as the point  $P(x, y)$  with co-ordinates  $x, y$  is called the complex plane. The horizontal  $x$ -axis is called real axis and vertical  $y$ -axis is called imaginary axis.

eg:



The number  $3 + 2i$  in the complex plane.

## Polar form of Complex number:

Let  $z = x + iy$  be a non-zero complex number represented by the point  $P(x, y)$  in the complex plane.

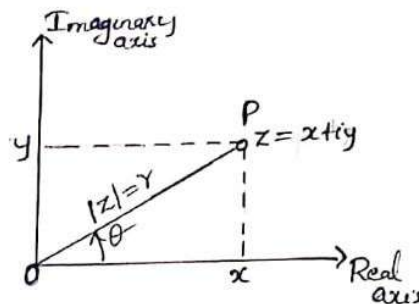
Then the polar co-ordinates  $(r, \theta)$  of  $P(x, y)$  defined by

$$x = r \cos \theta, \quad y = r \sin \theta$$

( $\theta$  is the directed angle from positive  $x$ -axis to  $OP$  &  $OP = r$ )

$$\begin{aligned} \text{Then } z &= x + iy \\ &= r \cos \theta + i r \sin \theta \\ &= r (\cos \theta + i \sin \theta) \end{aligned}$$

$$\text{i.e., } \boxed{z = r e^{i\theta}, \quad r > 0}$$



is called polar form of the complex number  $z = x + iy$

$r = \sqrt{x^2 + y^2} = |z|$  is called absolute value or modulus of  $z$ .

$\theta = \tan^{-1}(y/x) = \arg(z)$  is called argument of  $z$ .

Note: The value of  $\theta$  which satisfies  $-\pi \leq \theta \leq \pi$  is called the principal value of  $\theta$  or the principle argument of  $z$ , denoted by  $\text{Arg}(z)$

Complex Conjugate: The complex conjugate  $\bar{z}$  of a complex number  $z = x + iy$  is defined by

$$\bar{z} = x - iy$$

### Properties

$$\text{i) } |\bar{z}| = |z|$$

$$\text{iii) } \text{Re}(z) = x = \frac{z + \bar{z}}{2}$$

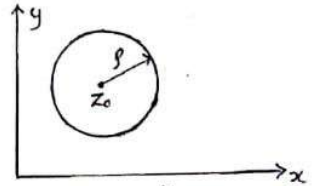
$$\text{ii) } |z|^2 = z \bar{z}$$

$$\text{iv) } \text{Im}(z) = y = \frac{z - \bar{z}}{2i}$$

## BASIC DEFINITIONS :

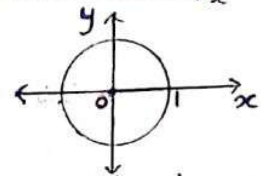
1. Circle : The general equation of a circle with centre ' $z_0$ ' and radius ' $\rho$ ' is

$$|z - z_0| = \rho$$



Note : The unit circle is given by

$$|z| = 1 ; \text{ centre } 0 \text{ and radius } 1.$$



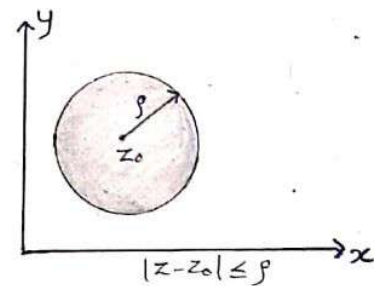
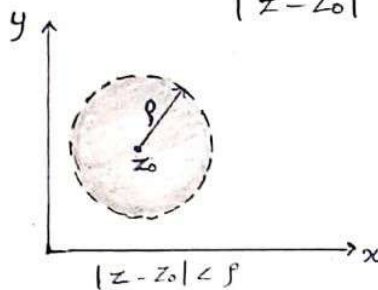
2. Open Circular disc : The set of all points lying inside the circle (but not on the circle). It is given by

$$|z - z_0| < \rho$$

This is also called the neighborhood of  $z_0$ .

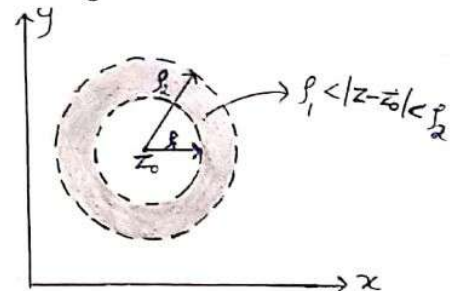
3. Closed Circular disc : The set of all points lying inside and on the circle. It is given by

$$|z - z_0| \leq \rho$$



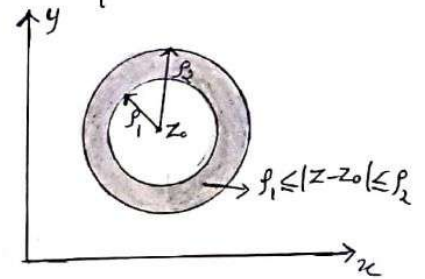
4. Open annulus (Circular ring) : The set of all points ' $z$ ' whose distance from  $z_0$  (i.e.,  $|z - z_0|$ ) is greater than  $\rho_1$  and less than  $\rho_2$ . It is given by

$$\rho_1 < |z - z_0| < \rho_2$$



Closed annulus : The set of all points whose distance from  $z_0$  (i.e.  $|z-z_0|$ ) is greater than or equal to  $\rho_1$  and less than or equal to  $\rho_2$ . It is given by

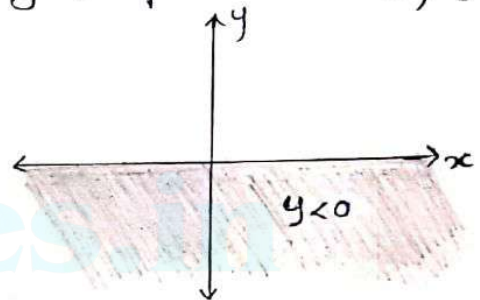
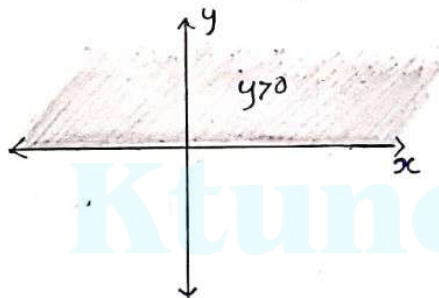
$$\rho_1 \leq |z-z_0| \leq \rho_2$$



Half Planes :

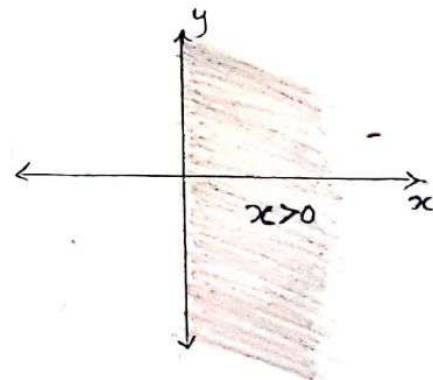
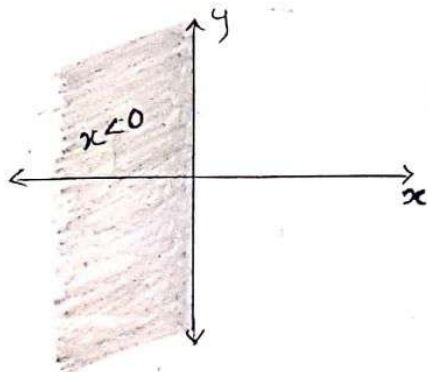
Upper half plane : The set of all points  $z=x+iy$ ;  $y \geq 0$

Lower half plane : The set of all points  $z=x+iy$ ;  $y \leq 0$



Left half plane : The set of all points  $z=x+iy$ ;  $x \leq 0$

Right half plane : The set of all points  $z=x+iy$ ;  $x \geq 0$



Problems: Determine and sketch the regions in the complex plane.

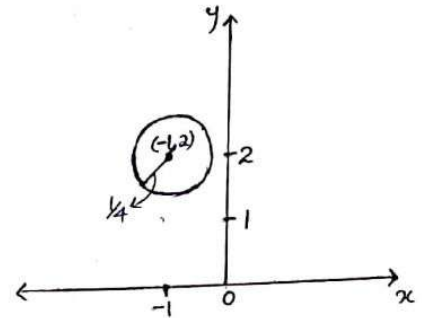
1)  $|z+1-2i| \leq 1/4$

Ans:  $|z+1-2i| = |z-(-1+2i)| \leq 1/4$

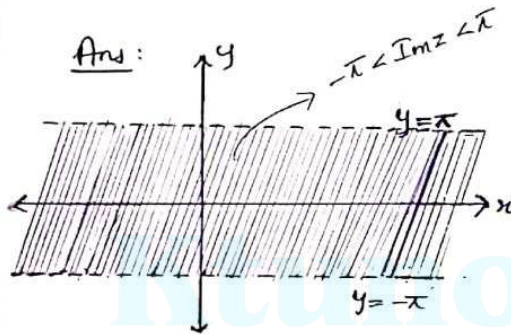
(of the form  $|z-z_0| \leq \rho$ )

$z_0 = -1+2i \rightarrow \text{centre}$

$\rho = 1/4 \rightarrow \text{radius}$



2)  $-\pi < \text{Im } z < \pi$



3)  $|z+i| \geq |z-i|$

Ans:  $|z+i| \geq |z-i|$

$\Rightarrow |x+iy+i| \geq |x+iy-i|$

$\Rightarrow |x+i(y+1)| \geq |x+i(y-1)|$

$\Rightarrow \sqrt{x^2+(y+1)^2} \geq \sqrt{x^2+(y-1)^2}$

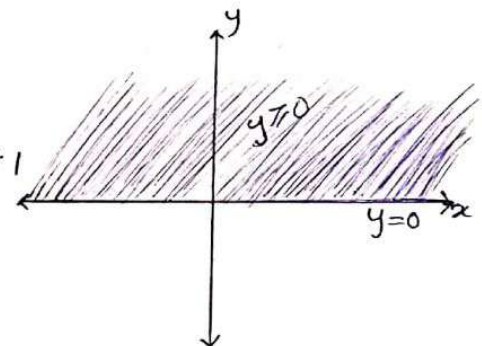
$\Rightarrow x^2+(y+1)^2 \geq x^2+(y-1)^2$

$\Rightarrow x^2+y^2+2y+1 \geq x^2+y^2-2y+1$

$\Rightarrow 2y \geq -2y$

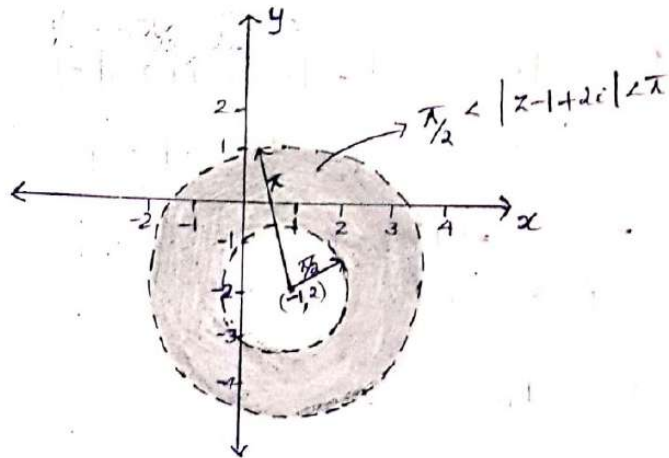
$\Rightarrow 4y \geq 0$

$y \geq 0$

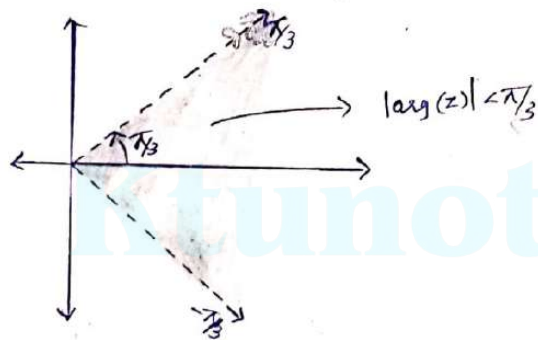


4)  $\pi/2 < |z-1+2i| < \pi$

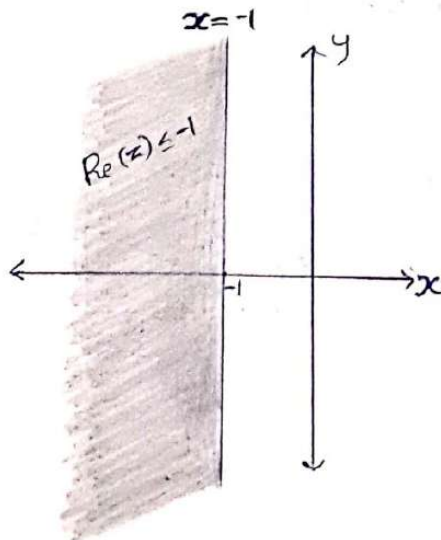
Ans:  $\rho_1 < |z-z_0| < \rho_2 \quad \therefore \rho_1 = \pi/2, z_0 = 1-2i; \rho_2 = \pi$



5)  $|\arg(z)| < \pi/3$



6)  $\operatorname{Re}(z) \leq -1$



## Complex Function

Let  $S$  be a set of complex numbers.

A function ' $f$ ' defined on  $S$  is a rule that assigns to every ' $z$ ' in  $S$  a complex number ' $w$ ', called the value of  $f$  at  $z$ . We write it as

$$w = f(z)$$

Here ' $z$ ' varies in  $S$  and is called a complex variable.

The set  $S$  is called the domain of  $f$ .

The set of all values of a function  $f$  is called the range of  $f$ .

Since  $w$  is a complex number, we write  $w = u + iv$ , where  $u$  and  $v$  are the real and imaginary parts respectively.

Since  $w$  depends on  $z = x + iy$ , ' $u$ ' becomes a real function of ' $x$ ' and ' $y$ '. ' $v$ ' also is a real function of  $x$  and  $y$ . Thus we write

$$w = f(z) = u(x, y) + iv(x, y)$$

Example : Let  $w = f(z) = z^2 + 3z$ . Find  $u$  and  $v$ .

Calculate the value of  $f$  at  $z = 1 + 3i$ .

Answer :

$$\begin{aligned} f(z) &= z^2 + 3z \\ &= (x + iy)^2 + 3(x + iy) \\ &= x^2 + 2ixy - y^2 + 3x + i3y \\ &= (x^2 - y^2 + 3x) + i(2xy + 3y) \\ &= u + iv \end{aligned}$$

$$\therefore u = x^2 - y^2 + 3x, \quad v = 2xy + 3y$$

$$f/z=1+3i = (1+3i)^2 + 3(1+3i) = 1 + 6i - 9 + 3 + 9i = 15i - 5$$

2) Let  $w = f(z) = 2iz + 6\bar{z}$ . Find  $u$  and  $v$  and the value of  $f$  at  $z = \frac{1}{2} + 4i$ .

Ans:  $f(z) = 2iz + 6\bar{z} = 2i(x+iy) + 6(x-iy)$   
 $= 2ix - 2y + 6x - 6iy$   
 $= (6x - 2y) + i(2x - 6y)$

$\therefore u(x, y) = 6x - 2y, v(x, y) = 2x - 6y$

$f(\frac{1}{2} + 4i) = 2i(\frac{1}{2} + 4i) + 6(\frac{1}{2} - 4i) = -5 - 23i$

Q) Find  $\text{Re} f$ ,  $\text{Im} f$  and their values at the given point  $z$ .

i)  $f(z) = 5z^2 - 12z + 3 + 2i$  at  $4 - 3i$

ii)  $f(z) = \frac{1}{1+z}$  at  $1 - i$     iii)  $f(z) = \frac{z-1}{z+1}$  at  $2i$

Ans:

i)  $f(z) = 5(x+iy)^2 - 12(x+iy) + 3 + 2i$   
 $= 5(x^2 - y^2 + 2ixy) - 12x - i12y + 3 + 2i$   
 $= (5x^2 - 5y^2 - 12x + 3) + i(10xy - 12y + 2)$

$\therefore \text{Re} f = 5x^2 - 5y^2 - 12x + 3, \text{Im} f = 10xy - 12y + 2$

$\text{Re} f / 4-3i = -10$  (putting  $x=4, y=-3$ )  $\text{Im} f / 4-3i = -82$

ii)  $f(z) = \frac{1}{1+z} = \frac{1}{1+(x+iy)} = \frac{1}{(1+x)+iy} = \frac{(1+x)-iy}{(1+x)^2+y^2}$   
 $= \frac{1+x}{(1+x)^2+y^2} + i \left( \frac{-y}{(1+x)^2+y^2} \right)$

$\therefore \text{Re} f = \frac{1+x}{(1+x)^2+y^2}, \text{Im} f = \frac{-y}{(1+x)^2+y^2}$

$\text{Re} f / 1-i = \frac{1+1}{(1+1)^2+1^2} = \frac{2}{5}, \text{Im} f / 1-i = \frac{-(-1)}{(1+1)^2+1^2} = \frac{1}{5}$

iii)  $f(z) = \frac{z-1}{z+1} = \frac{(x+iy)-1}{(x+iy)+1} = \frac{(x-1)+iy}{(x+1)+iy} = \frac{[(x-1)+iy][(x+1)-iy]}{[(x+1)+iy][(x+1)-iy]}$   
 $= \frac{(x^2-1)+y^2+2iy}{(x+1)^2+y^2} = \frac{x^2+y^2-1}{(x+1)^2+y^2} + i \frac{2y}{(x+1)^2+y^2}$   
 $\therefore \text{Re} f = \frac{x^2+y^2-1}{(x+1)^2+y^2}, \text{Im} f = \frac{2y}{(x+1)^2+y^2} \Rightarrow \text{Re} f / 2i = \frac{3}{5}, \text{Im} f / 2i = \frac{4}{5}$

## Limit, Continuity & Derivative of a complex function

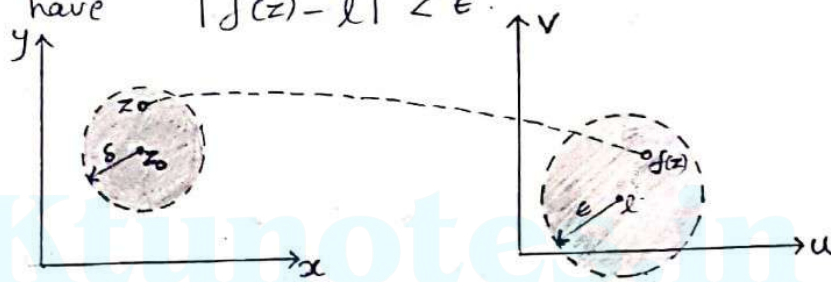
Limit: A function  $f(z)$  is said to have the limit ' $l$ ' as ' $z$ ' approaches a point ' $z_0$ ' written

$$\lim_{z \rightarrow z_0} f(z) = l$$

if ' $f$ ' is defined in a neighbourhood of ' $z_0$ ' and if the values of ' $f$ ' are close to ' $l$ ' for all ' $z$ ' close to ' $z_0$ '

OR

If for every positive real ' $\epsilon$ ' we can find a positive real ' $\delta$ ' such that for all  $z \neq z_0$  in the disk  $|z - z_0| < \delta$  we have  $|f(z) - l| < \epsilon$ .



Limit in the complex plane.

Note: (i)  $z \rightarrow z_0$  in the complex plane means that ' $z$ ' approaches ' $z_0$ ' from any direction along any straight line or a curved path and if the limit exists at  $z_0$ , then it must be same along all these paths

(ii) If we obtain two different limits as  $z \rightarrow z_0$  along two different paths, we conclude that limit does not exist.

For finding limit doesn't exist at  $z=0$  we usually consider the following paths

Path I:  $z \rightarrow 0$  along x-axis &  $z \rightarrow 0$  along y-axis

Path II:  $z \rightarrow 0$  along a straight line through origin  $y = mx$

Path III:  $z \rightarrow 0$  along  $y = mx^2$

Continuity: A function  $f(z)$  is said to be continuous at  $z = z_0$  if  $f(z_0)$  is defined and  $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

Derivative: The derivative of a complex function  $f'$  at a point  $z_0$  written  $f'(z_0)$  and is defined by

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

provided the limit exists.

Then  $f'$  is said to be differentiable at  $z_0$ .

If we write  $\Delta z = z - z_0$  then

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

[Differentiability at  $z_0$  means that along whatever path  $z$  approaches  $z_0$ , the above limit should be the same. The limit  $f'(z_0)$  is known as the derivative of  $f(z)$  at  $z_0$ ]

Note: The differentiation rules are the same as in real calculus. For any differentiable functions  $f$  and  $g$  and constant  $c$ , we have

i)  $(cf)' = cf'$

ii)  $(f+g)' = f' + g'$

iii)  $(fg)' = f'g + g'f$

iv)  $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

v)  $(z^n)' = nz^{n-1}$ ,  $n$  integer

vi) Chain rule

## Problems

1) Find the limits of the following

i)  $\lim_{z \rightarrow 1} \frac{z^2 - 1}{z - 1}$       ii)  $\lim_{z \rightarrow 2i} (3z + iy^2)$

Ans: i)  $\lim_{z \rightarrow 1} \frac{z^2 - 1}{z - 1} = \lim_{z \rightarrow 1} \frac{(z-1)(z+1)}{z-1} = \lim_{z \rightarrow 1} (z+1) = \underline{2}$

ii) we have  $z = x + iy = 2i \Rightarrow x = 0, y = 2$

$\therefore \lim_{z \rightarrow 2i} (3z + iy^2) = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 2}} (3x + iy^2) = 0 + i2^2 = \underline{4i}$

2) Prove that the limit  $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$  does not exist.

Ans: We shall show that as  $z \rightarrow 0$  along two different paths, we obtain two different limits & therefore the limit does not exist.

Path (i): Let  $z \rightarrow 0$  along the  $x$ -axis, then  $y = 0$

$\therefore z = x + iy = x$  and  $\bar{z} = x - iy = x$ , so that

$$\lim_{z \rightarrow 0} \frac{\bar{z}}{z} = \lim_{x \rightarrow 0} \frac{x}{x} = \underline{1}$$

Path (ii): Let  $z \rightarrow 0$  along the  $y$ -axis, then  $x = 0$

$\therefore z = x + iy = iy$  and  $\bar{z} = x - iy = -iy$

$\therefore \lim_{z \rightarrow 0} \frac{\bar{z}}{z} = \lim_{y \rightarrow 0} \frac{-iy}{iy} = \underline{-1}$

Since the limit is not unique, the limit  $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$  doesn't exist

**OR**

[Consider any straight line path  $y = mx$

$$\begin{aligned} \therefore \lim_{z \rightarrow 0} \frac{\bar{z}}{z} &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x - iy}{x + iy} = \lim_{x \rightarrow 0} \frac{x - imx}{x + imx} \\ &= \lim_{x \rightarrow 0} \frac{x(1 - im)}{x(1 + im)} = \frac{1 - im}{1 + im} \end{aligned}$$

The limit depends on 'm' and so it is not unique  
So the limit  $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$  does not exist]

3) Check whether the function  $f(z)$  is continuous at  $z=0$

$$\text{if i) } f(z) = \begin{cases} |z|^2 \operatorname{Im}\left(\frac{1}{z}\right) & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$$

$$\text{ii) } f(z) = \begin{cases} \frac{\operatorname{Im}(z^2)}{|z|^2} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$$

$$\text{iii) } f(z) = \begin{cases} \operatorname{Re}\left(\frac{z^2}{|z|}\right) & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$$

Ans: A function  $f(z)$  is continuous at  $z=0$  if  

$$\lim_{z \rightarrow 0} f(z) = f(0)$$

i) Here  $f(0) = 0$  (given)

$$\begin{aligned} \text{Consider } \lim_{z \rightarrow 0} f(z) &= \lim_{z \rightarrow 0} |z|^2 \operatorname{Im}\left(\frac{1}{z}\right) \\ &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} (x^2 + y^2) \left( \frac{-y}{x^2 + y^2} \right) \\ &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} -y = 0 \end{aligned}$$

$$\therefore \lim_{z \rightarrow 0} f(z) = 0 = f(0)$$

Hence  $f(z)$  is continuous at  $z=0$

$$\begin{aligned} z &= x+iy \\ \Rightarrow |z| &= \sqrt{x^2+y^2} \\ \Rightarrow \frac{1}{z} &= \frac{1}{x+iy} \\ &= \frac{x-iy}{(x+iy)(x-iy)} \\ &= \frac{x-iy}{x^2+y^2} \\ \therefore \operatorname{Im}\left(\frac{1}{z}\right) &= \frac{-y}{x^2+y^2} \end{aligned}$$

ii)  $f(0) = 0$  (given)

$$\begin{aligned} \lim_{z \rightarrow 0} f(z) &= \lim_{z \rightarrow 0} \frac{\operatorname{Im}(z^2)}{|z|^2} \\ &= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{2xy}{x^2+y^2} \end{aligned}$$

Let  $z \rightarrow 0$  along the path  $y = mx$

$$\therefore \lim_{z \rightarrow 0} f(z) = \lim_{x \rightarrow 0} \frac{2x \cdot mx}{x^2 + m^2 x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2mx^2}{x^2(1+m^2)} = \lim_{x \rightarrow 0} \frac{2m}{1+m^2} = \frac{2m}{1+m^2}$$

$$\begin{aligned} z &= x+iy \\ z^2 &= (x+iy)^2 \\ &= x^2 + 2ixy - y^2 \\ &= x^2 - y^2 + i2xy \\ \therefore \operatorname{Im}(z^2) &= 2xy \end{aligned}$$

The value of the limit depends on 'm' and so it is not unique

$\therefore \lim_{z \rightarrow 0} f(z)$  doesn't exist at  $z=0$ .

Hence  $f(z)$  is not continuous at  $z=0$ .

ii)  $f(0) = 0$  &  $f(z) = \operatorname{Re}\left(\frac{z^2}{|z|}\right)$  for  $z \neq 0$

Along the path  $y=mx$

$$\lim_{z \rightarrow 0} f(z) = \lim_{z \rightarrow 0} \operatorname{Re}\left(\frac{z^2}{|z|}\right)$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 - y^2}{\sqrt{x^2 + y^2}}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 - m^2 x^2}{\sqrt{x^2 + m^2 x^2}}$$

$$= \lim_{x \rightarrow 0} \frac{x^2(1-m^2)}{x\sqrt{1+m^2}} = \lim_{x \rightarrow 0} \frac{x(1-m^2)}{\sqrt{1+m^2}} = 0$$

$$\begin{aligned} z &= x+iy \\ \Rightarrow z^2 &= x^2 - y^2 + i2xy \\ &\& |z| = \sqrt{x^2 + y^2} \\ \therefore \operatorname{Re}\left(\frac{z^2}{|z|}\right) &= \operatorname{Re}\left[\frac{(x^2 - y^2 + i2xy)}{\sqrt{x^2 + y^2}}\right] \\ &= \frac{x^2 - y^2}{\sqrt{x^2 + y^2}} \end{aligned}$$

i.e.,  $\lim_{z \rightarrow 0} f(z) = 0$  along the path  $y=mx$

$\Rightarrow \lim_{z \rightarrow 0} f(z) = 0$  when  $z \rightarrow 0$  along any path in the complex plane.

$\therefore$  we should not conclude that the limit exists and hence  $f(z)$  is continuous at  $z=0$ .

$\therefore$  to test the continuity, we consider the polar form of  $f(z)$

Putting  $x=r \cos \theta$ ,  $y=r \sin \theta$

$$\therefore z = x+iy = re^{i\theta} \Rightarrow z^2 = (re^{i\theta})^2 = r^2 e^{i2\theta} \quad \& \quad |z| = r$$

$$\begin{aligned} \therefore f(z) &= \operatorname{Re}\left(\frac{z^2}{|z|}\right) = \operatorname{Re}\left(\frac{r^2 e^{i2\theta}}{r}\right) = \operatorname{Re}(re^{i2\theta}) \\ &= \operatorname{Re}(r(\cos 2\theta + i \sin 2\theta)) \\ &= r \cos 2\theta \end{aligned}$$

When  $z \rightarrow 0$ ,  $r \rightarrow 0$

$$\therefore \lim_{z \rightarrow 0} f(z) = \lim_{r \rightarrow 0} r \cos 2\theta = 0 = f(0)$$

$\Rightarrow$   $f(z)$  is continuous at  $z=0$ .

4) Show that  $f(z) = \begin{cases} \frac{z \operatorname{Re}(z)}{|z|} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$  is not differentiable at  $z = 0$ .

Ans : A function  $f(z)$  is differentiable at  $z = 0$

if  $\lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z}$  exists.

Here  $f(z) = \frac{z \operatorname{Re}(z)}{|z|} = \frac{(x+iy)x}{\sqrt{x^2+y^2}} ; f(0) = 0$

$$\therefore \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left[ \frac{\frac{(x+iy)x}{\sqrt{x^2+y^2}} - 0}{x+iy} \right] = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \left[ \frac{(x+iy)x}{\sqrt{x^2+y^2} (x+iy)} \right]$$

Let  $z \rightarrow 0$  along the path  $y = mx$

$$\begin{aligned} \therefore \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} &= \lim_{x \rightarrow 0} \frac{(x+imx)x}{\sqrt{x^2+m^2x^2} (x+imx)} \\ &= \lim_{x \rightarrow 0} \frac{x}{\sqrt{x^2+m^2x^2}} = \lim_{x \rightarrow 0} \frac{x}{x\sqrt{1+m^2}} \\ &= \frac{1}{\sqrt{1+m^2}}, \text{ not unique (depends on 'm')} \end{aligned}$$

So limit does not exist at  $z = 0$ .

$\therefore f(z)$  is not differentiable at  $z = 0$

5) If  $f(z)$  is differentiable at  $z_0$ , show that  $f(z)$  is continuous at  $z_0$ .

Ans : Suppose  $f$  is differentiable at a point  $z_0$

Then  $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$  exists and is equal to  $f'(z_0)$ .

$$\begin{aligned} \text{we write } \lim_{z \rightarrow z_0} [f(z) - f(z_0)] &= \lim_{z \rightarrow z_0} \left[ \frac{f(z) - f(z_0)}{z - z_0} (z - z_0) \right] \\ &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \cdot \lim_{z \rightarrow z_0} (z - z_0) \end{aligned}$$

$$= f'(z_0) \cdot \lim_{z \rightarrow z_0} f(z) = 0$$

$$\text{i.e., } \lim_{z \rightarrow z_0} [f(z) - f(z_0)] = 0$$

$$\therefore \lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} f(z_0) = f(z_0)$$

$\therefore f(z)$  is continuous at  $z_0$ .

Note : Converse is not true. i.e., a function may be continuous but not differentiable.

6) Show that  $f(z) = \text{Re}(z) = x$  is continuous but not differentiable

Ans: Continuity : For any point  $z$ ,

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} x = \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} x = x_0 = f(z_0)$$

Differentiability : For any  $z$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{(x + \Delta x) - x}{\Delta x + i \Delta y}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{\Delta x}{\Delta x + i \Delta y}$$

We consider two paths for  $\Delta z \rightarrow 0$

Path I : Let  $\Delta y \rightarrow 0$  first and then  $\Delta x \rightarrow 0$

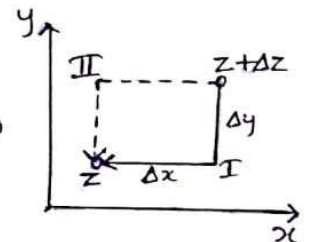
$$\therefore f'(z) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta x}{\Delta x + i \Delta y} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

Path II : Let  $\Delta x \rightarrow 0$  first and then  $\Delta y \rightarrow 0$

$$\therefore f'(z) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\Delta x}{\Delta x + i \Delta y} = 0$$

So limit doesn't exist.

i,  $f(z)$  is not differentiable.



a) Using differentiation rules find the derivative of following functions at given points.

i)  $f(z) = (z-2i)^3$  at  $z = 5+2i$

ii)  $f(z) = \frac{1.5z+2i}{3iz-4}$  at any  $z$

iii)  $f(z) = i(1-z)^n$  at  $0$

iv)  $f(z) = \frac{z^3}{(z-i)^3}$  at  $-i$

Ans : i)  $f'(z) = 3(z-2i)^2$  (using power rule:  $z^n = n z^{n-1}$ )

$\therefore f'(5+2i) = 3(5+2i-2i)^2 = \underline{\underline{75}}$

ii)  $f(z) = \frac{1.5z+2i}{3iz-4}$

$f'(z) = \frac{(3iz-4)(1.5) - (1.5z+2i)(3i)}{(3iz-4)^2}$

$= \frac{4.5iz - 6 - 4.5iz + 6}{(3iz-4)^2} = 0$

$\therefore f'(z) = 0$  at any  $z$

iii)  $f(z) = i(1-z)^n$

$f'(z) = i n (1-z)^{n-1} \frac{d}{dz}(1-z)$

$= i n (1-z)^{n-1} (-1)$

$\therefore f'(0) = \underline{\underline{-in}}$

iv)  $f(z) = \frac{z^3}{(z-i)^3}$

$f'(z) = \frac{(z-i)^3 3z^2 - z^3 \cdot 3(z-i)^2 \cdot 1}{(z-i)^6}$

$= \frac{z^2(z-i)^2 [(z-i)3 - 3z]}{(z-i)^6} = \frac{z^2 [(z-i)3 - 3z]}{(z-i)^4}$

$\therefore f'(-i) = \frac{(-i)^2 [(-i-i)3 - 3(-i)]}{(-i-i)^4} = \frac{-1 [-6i + 3i]}{(-2i)^4}$

$= \underline{\underline{\frac{3i}{16}}}$

## Home Work

1. Determine and sketch or graph the sets in the complex plane

i)  $0 < |z| < 1$       ii)  $\operatorname{Re}(\frac{1}{z}) < 1$

2. Test the continuity at  $z=0$  if

i)  $f(z) = \begin{cases} \frac{\operatorname{Im}(z)}{|z|} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$

ii)  $f(z) = \begin{cases} \frac{\operatorname{Re}(z^2)}{|z|^2} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$

iii)  $f(z) = \begin{cases} \frac{\operatorname{Re}(z)}{1-|z|} & ; z \neq 0 \\ 0 & ; z = 0 \end{cases}$

3. Find the value of the derivative of

i)  $\frac{z-i}{z+i}$  at  $i$

ii)  $(iz^3 + 3z^2)^3$  at  $2i$

## Analytic Functions

A function  $f(z)$  is said to be analytic at a point  $z = z_0$  if it is analytic in a neighbourhood of  $z_0$ . In other words,  $f$  is analytic at a point  $z_0$  if its derivative exists not only at  $z_0$  but also at each point  $z$  in some neighbourhood of  $z_0$ .

The function  $f(z)$  is said to be analytic in a domain  $D$  if  $f(z)$  is defined and differentiable at all points of  $D$ .

Sometimes analytic functions are also called holomorphic or regular functions. Also by an analytic function we mean a function that is analytic in some domain.

[\* Domain : An open connected set is called a domain.

A set  $S$  is called open if every point of  $S$  has a neighbourhood consisting entirely of points that belong to  $S$ . and a set  $S$  is called connected if any two of its points can be joined by a chain of finitely many straight line segments all of whose points belong to  $S$ .]

Note :

i) Non-negative integer powers  $1, z, z^2, \dots$  are analytic in the entire complex plane.

ii) Polynomial functions of the form  $c_0 + c_1 z + \dots + c_n z^n$  are also analytic in the entire complex plane.

iii) The quotient of two polynomial functions  $g(z)$  and  $h(z)$ ,

$$f(z) = \frac{g(z)}{h(z)}$$
, called a rational function is analytic except at the points where  $h(z) = 0$

## Cauchy - Riemann Equations [ C-R equations: Necessary part ]

Theorem: Let  $f(z) = u(x, y) + iv(x, y)$  be defined and continuous in some neighbourhood of a point  $z = x + iy$  and differentiable at  $z$  itself. Then, at that point, the first order partial derivatives of  $u$  and  $v$  exist and satisfy the Cauchy - Riemann Equations

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad \text{--- (1)}$$

Hence, if  $f(z)$  is analytic in a domain  $D$ , these partial derivatives exist and satisfy (1) at all points of  $D$ .

Proof:

Given that  $f(z) = u(x, y) + iv(x, y)$  is differentiable at  $z$ , i.e., the derivative  $f'(z)$  at  $z$  exists.

$$\text{i.e., } f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \text{ exists.} \quad \text{--- (2)}$$

We write  $z = x + iy$  and  $\Delta z = \Delta x + i\Delta y$

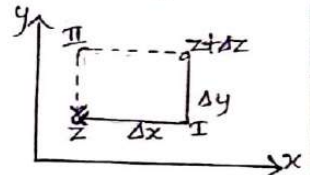
$$\text{Then } z + \Delta z = (x + iy) + (\Delta x + i\Delta y) = (x + \Delta x) + i(y + \Delta y)$$

In terms of  $u$  and  $v$ ,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{[u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)] - [u(x, y) + iv(x, y)]}{\Delta x + i\Delta y} \quad \text{--- (3)}$$

We consider two paths for  $\Delta z \rightarrow 0$

Path I: Let  $\Delta y \rightarrow 0$  first and then  $\Delta x \rightarrow 0$



$$\text{(3)} \Rightarrow f'(z) = \lim_{\substack{\Delta y \rightarrow 0 \\ \Delta x \rightarrow 0}} \frac{u(x + \Delta x, y + \Delta y) - u(x, y)}{\Delta x + i\Delta y} + \lim_{\substack{\Delta y \rightarrow 0 \\ \Delta x \rightarrow 0}} \frac{i[v(x + \Delta x, y + \Delta y) - v(x, y)]}{\Delta x + i\Delta y}$$

After  $\Delta y$  is zero

$$= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{i[v(x + \Delta x, y) - v(x, y)]}{\Delta x}$$

Since  $f'(z)$  exists, the limits on the right exist and give the partial derivatives of  $u$  and  $v$  w.r.t 'x'

$$\therefore f'(z) = u_x + i v_x \quad \text{--- (4)}$$

Path II: Let  $\Delta x \rightarrow 0$  first and then  $\Delta y \rightarrow 0$

$$\begin{aligned} (3) \Rightarrow f'(z) &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{u(x+\Delta x, y+\Delta y) - u(x, y)}{\Delta x + i \Delta y} + \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} i \frac{v(x+\Delta x, y+\Delta y) - v(x, y)}{\Delta x + i \Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{u(x, y+\Delta y) - u(x, y)}{i \Delta y} + \lim_{\Delta y \rightarrow 0} i \frac{v(x, y+\Delta y) - v(x, y)}{i \Delta y} \end{aligned}$$

Since  $f'(z)$  exists, the limits on the right exist and give the partial derivatives of  $u$  and  $v$  w.r.t 'y'

$$\therefore f'(z) = \frac{1}{i} u_y + v_y = -i u_y + v_y \quad [\because \frac{1}{i} = \frac{i}{i^2} = -i] \quad \text{--- (5)}$$

The existence of the derivative  $f'(z)$  implies the existence of the four partial derivatives  $u_x, u_y, v_x$  &  $v_y$  in (4) & (5).

Since the limit in (2) exists, the limit value along paths I & II must be equal.

$$\therefore (4) = (5) \Rightarrow u_x + i v_x = -i u_y + v_y$$

Equating the real & imaginary parts, we get

$$u_x = v_y \quad \& \quad u_y = -v_x$$

This proves the first statement of the theorem.

If  $f(z)$  is analytic in a domain  $D$ , then  $f(z)$  is defined and differentiable at all points of  $D$ .

$\Rightarrow$  partial derivatives  $u_x, u_y, v_x, v_y$  exist & satisfy C-R eqns at all points of  $D$

Hence the second statement of the theorem.

Note 1: Derivative  $f'(z)$  can be calculated using formulas (4) & (5)

$$(4) \Rightarrow f'(z) = u_x + i v_x = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$(5) \Rightarrow f'(z) = v_y - i u_y = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Note 2: C-R equations are necessary but not sufficient to ensure the existence of the derivative of a function  $f(z)$  at a point. But with certain continuity conditions, we have the following theorem.

Theorem (C-R equations - Sufficiency part)

A function  $f(z) = u(x, y) + i v(x, y)$  (where  $u(x, y)$  &  $v(x, y)$  are real valued continuous functions) is differentiable at a point, if the four partial derivatives  $u_x, u_y, v_x, v_y$  are continuous and satisfy Cauchy-Riemann equations at that point.

Hence if two real valued continuous functions  $u(x, y)$  &  $v(x, y)$  have continuous first partial derivatives that satisfy C-R equations in some domain  $D$ , then the complex function  $f(z) = u(x, y) + i v(x, y)$  is analytic in  $D$ .

Cauchy-Riemann Equations in Polar form

In polar form  $z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$

Set  $f(z) = u(r, \theta) + i v(r, \theta)$

Then the Cauchy-Riemann equations are

$$u_r = \frac{1}{r} v_\theta ; v_r = -\frac{1}{r} u_\theta ; r > 0$$

Entire function: A function which is analytic for all  $z$  in the complex plane is known as entire function

### Problems :

1. Show that  $f(z) = |z|^2$  is differentiable only at  $z=0$ .

Ans: Let  $f(z) = |z|^2 = x^2 + y^2 = u + iv$

$$\Rightarrow u = x^2 + y^2 ; v = 0$$

$$\therefore u_x = \frac{\partial u}{\partial x} = 2x ; v_x = \frac{\partial v}{\partial x} = 0$$

$$u_y = \frac{\partial u}{\partial y} = 2y ; v_y = \frac{\partial v}{\partial y} = 0$$

Here the partial derivatives are continuous at all points  $x$  and  $y$ . But the C-R equations are satisfied only at  $x=0$  and  $y=0$ .

$$\text{i.e., } u_x = v_y \text{ \& } u_y = -v_x \text{ only at } (0,0)$$

$\therefore f(z)$  is differentiable only at  $(0,0)$  i.e., at  $z=0$ .

Note:  $f'(z)$  doesn't exist at any non-zero point.  
 $\Rightarrow f(z)$  is not analytic anywhere.

2. Using C-R equations, show that  $f(z) = \bar{z}$  is not differentiable at any point ( $\Rightarrow f(z)$  is nowhere analytic).

Ans: Let  $f(z) = \bar{z} = x - iy = u + iv$

$$\Rightarrow u = x , v = -y$$

$$\therefore u_x = \frac{\partial u}{\partial x} = 1 ; v_x = \frac{\partial v}{\partial x} = 0$$

$$u_y = \frac{\partial u}{\partial y} = 0 ; v_y = \frac{\partial v}{\partial y} = -1$$

Here the partial derivatives are continuous everywhere.

But C-R equations are not satisfied at any point.

Thus the function is not differentiable at any point.

$\Rightarrow f(z)$  is nowhere analytic.

3. Using C-R equations, show that  $f(z) = \frac{1}{z}$ ,  $z \neq 0$  is analytic except at the point  $z=0$ .

Ans: Let  $f(z) = \frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)} = \frac{x-iy}{x^2+y^2}$

$$\therefore u = \frac{x}{x^2+y^2}, \quad v = \frac{-y}{x^2+y^2}$$

$$u_x = \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left( \frac{x}{x^2+y^2} \right) = \frac{(x^2+y^2) \cdot 1 - x \cdot 2x}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

$$u_y = \frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left( \frac{x}{x^2+y^2} \right) = \frac{(x^2+y^2) \cdot 0 - x \cdot 2y}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

$$v_x = \frac{\partial v}{\partial x} = \frac{\partial}{\partial x} \left( \frac{-y}{x^2+y^2} \right) = \frac{(x^2+y^2)(0) + y \cdot 2x}{(x^2+y^2)^2} = \frac{2xy}{(x^2+y^2)^2}$$

$$v_y = \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \left( \frac{-y}{x^2+y^2} \right) = \frac{(x^2+y^2)(-1) + y \cdot 2y}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

Here  $u_x = v_y$  &  $u_y = -v_x$  except at  $(0,0)$

$\therefore$  C-R equations are satisfied at all points except at  $z=0$  and the partial derivatives are also continuous at all points except at  $z=0$ .

$\Rightarrow f(z) = \frac{1}{z}$  is analytic at all points except at  $z=0$   
(by sufficiency part of analyticity)

4. Prove that  $f(z) = e^z$  is analytic. Find  $f'(z)$

Ans:

$$f(z) = e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x (\cos y + i \sin y)$$

$$\Rightarrow u + iv = e^x \cos y + i e^x \sin y \quad [e^{i\theta} = \cos \theta + i \sin \theta]$$

$$\therefore u = e^x \cos y, \quad v = e^x \sin y$$

$$\frac{\partial u}{\partial x} = e^x \cos y, \quad \frac{\partial v}{\partial x} = e^x \sin y$$

$$\frac{\partial u}{\partial y} = -e^x \sin y, \quad \frac{\partial v}{\partial y} = e^x \cos y$$

Hence  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  and  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Thus Cauchy - Riemann equations are satisfied for all  $z$   
 Also the first order partial derivatives are everywhere continuous ( $\because u_x, u_y, v_x, v_y$  are product of continuous functions)

Therefore  $f(z) = e^z$  is analytic for all  $z$   
 (by sufficient condition for analyticity)

$$\begin{aligned} \text{Also } f'(z) &= u_x + i v_x = e^x \cos y + i e^x \sin y \\ &= e^x (\cos y + i \sin y) \\ &= e^x \cdot e^{iy} \\ &= e^{x+iy} = \underline{\underline{e^z}} \end{aligned}$$

5. Prove that the function  $\sin z$  is analytic and find its derivative

Ans:

$$\begin{aligned} f(z) = \sin z &= \sin(x+iy) \\ &= \sin x \cos(iy) + \cos x \sin(iy) \\ &= \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

$$\left[ \begin{array}{l} \sin(A+B) = \sin A \cos B + \cos A \sin B \\ \text{Also } \cos(iy) = \cosh y \\ \sin(iy) = i \sinh y \end{array} \right\} \text{hyperbolic functions.}$$

$$\therefore f(z) = u + i v = \sin x \cosh y + i \cos x \sinh y$$

$$\Rightarrow u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\therefore \frac{\partial u}{\partial x} = \cos x \cosh y, \quad \frac{\partial v}{\partial x} = -\sin x \sinh y$$

$$\frac{\partial u}{\partial y} = \sin x \sinh y, \quad \frac{\partial v}{\partial y} = \cos x \cosh y$$

$$\left[ \begin{array}{l} \frac{d}{dx} (\sinh x) = \frac{d}{dx} \left( \frac{e^x - e^{-x}}{2} \right) = \frac{1}{2} (e^x + e^{-x}) = \cosh x \\ \frac{d}{dx} (\cosh x) = \frac{d}{dx} \left( \frac{e^x + e^{-x}}{2} \right) = \frac{1}{2} (e^x - e^{-x}) = \sinh x \end{array} \right]$$

$\therefore$  C-R equations are satisfied.

Also the partial derivatives  $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$  are continuous for every value  $x$  and  $y$ , since trigonometric and hyperbolic functions are continuous in their domains.

$\therefore$  by the sufficiency conditions for analyticity  $\sin z$  is analytic everywhere.

$$f'(z) = u_x + iv_x$$

$$= \cos x \cosh y - i \sin x \sinh y$$

$$= \cos x \cos(iy) - \sin x \sin(iy)$$

$$= \cos(x + iy)$$

$$= \cos z$$

$$\begin{cases} \cos A \cos B - \sin A \sin B \\ = \cos(A+B) \\ \sin(iy) = i \sinh y \end{cases}$$

$$\therefore \frac{d(\sin z)}{dz} = \cos z$$

6. Show that  $f(z) = e^{-x} \cos y - i e^{-x} \sin y$  is differentiable everywhere. Find its derivative.

Ans :  $f(z) = e^{-x} \cos y - i e^{-x} \sin y = u + iv$

$$\Rightarrow u = e^{-x} \cos y, \quad v = -e^{-x} \sin y$$

$$u_x = -e^{-x} \cos y, \quad v_x = e^{-x} \sin y$$

$$u_y = -e^{-x} \sin y, \quad v_y = -e^{-x} \cos y$$

$$\therefore u_x = v_y \quad \& \quad u_y = -v_x$$

$\therefore$  C-R equations are satisfied.

Also the partial derivatives  $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$  are continuous for every value  $x$  and  $y$ , since trigonometric and exponential functions are continuous.

$\therefore f(z)$  is differentiable everywhere.

$$f'(z) = u_x + iv_x = -e^{-x} \cos y + i e^{-x} \sin y = -e^{-x} = \underline{\underline{-\frac{1}{e^z}}}$$

7. Are the following functions analytic?

i)  $f(z) = iz\bar{z}$     ii)  $f(z) = \operatorname{Re}(z^2) - i\operatorname{Im}(z^2)$

iii)  $f(z) = \ln|z| + i\operatorname{Arg}(z)$     iv)  $f(z) = \operatorname{Arg}(z)$

Ans: i)  $f(z) = iz\bar{z} = i(x+iy)(x-iy) = i(x^2-y^2)$

ii,  $u+iv = i(x^2-y^2)$

$\Rightarrow u = 0$  ;  $v = x^2-y^2$

$\therefore u_x = 0$  ;  $v_x = 2x$

$u_y = 0$  ;  $v_y = -2y$

The four partial derivatives are continuous at all points  $x$  and  $y$ . But the C-R equations are satisfied only at  $x=0$  &  $y=0$ . Hence by the sufficiency conditions,  $f(z)$  is differentiable only at  $(0,0)$  i.e. at  $z=0$ .

ii,  $f(z)$  is not analytic at any point

or  $f(z)$  is nowhere analytic.

ii)  $f(z) = \operatorname{Re}(z^2) - i\operatorname{Im}(z^2)$

ii,  $u+iv = x^2-y^2 - i2xy$

$\Rightarrow u = x^2-y^2$  ;  $v = -2xy$

$\therefore u_x = 2x$  ;  $v_x = -2y$

$u_y = -2y$  ;  $v_y = -2x$

$$\begin{aligned} z &= x+iy \\ z^2 &= (x+iy)^2 \\ &= x^2-y^2+i2xy \end{aligned}$$

Here the partial derivatives are continuous at all points  $x$  and  $y$ . But the C-R equations are satisfied only at  $(0,0)$ .

$\therefore f(z)$  is differentiable only at  $(0,0)$

Hence  $f(z)$  is not analytic anywhere.

$$\text{iii) } f(z) = \ln|z| + i \operatorname{Arg}(z) = \ln \sqrt{x^2+y^2} + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$\Rightarrow u+iv = \ln \sqrt{x^2+y^2} + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$\therefore u = \ln \sqrt{x^2+y^2} ; v = \tan^{-1}\left(\frac{y}{x}\right)$$

$$u_x = \frac{\partial u}{\partial x} = \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2\sqrt{x^2+y^2}} \cdot 2x = \frac{x}{x^2+y^2}$$

$$u_y = \frac{\partial u}{\partial y} = \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2\sqrt{x^2+y^2}} \cdot 2y = \frac{y}{x^2+y^2}$$

$$v_x = \frac{\partial v}{\partial x} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{x^2}{x^2+y^2} \cdot \frac{-y}{x^2} = \frac{-y}{x^2+y^2}$$

$$v_y = \frac{\partial v}{\partial y} = \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x^2}{x^2+y^2} \cdot \frac{1}{x} = \frac{x}{x^2+y^2}$$

$\therefore$  C-R equations are satisfied at all points except  $z=0$ . Also the four partial derivatives are continuous for every value  $x$  and  $y$  except at  $z=0$ . But  $v = \tan^{-1}(y/x)$  is not continuous at all points on the negative real axis. Hence by the sufficiency conditions,  $f(z)$  is analytic everywhere in the complex plane except at  $0$  and on the negative real axis.

$$\text{iv) } f(z) = \operatorname{Arg}(z) = \theta$$

$$f(z) = u(r, \theta) + i v(r, \theta) = \theta$$

$$\text{ie, } u(r, \theta) = \theta, \quad v(r, \theta) = 0$$

Using polar form of C-R equations, we have

$$u_r = 0 ; v_r = 0$$

$$u_\theta = 1 ; v_\theta = 0$$

$$\therefore v_r \neq -\frac{1}{r} u_\theta \Rightarrow f(z) \text{ is not analytic.}$$

8. Are the following functions analytic?

$$\text{i) } f(z) = \frac{-i}{z^4} \quad \text{ii) } f(z) = \frac{3\pi^2}{z^3 + 4\pi^2 z} \quad \text{iii) } f(z) = \frac{1}{z - z^5}$$

Ans: The quotient  $f(z) = \frac{g(z)}{h(z)}$  is analytic everywhere except when  $h(z) = 0$

$$\text{i) } f(z) = \frac{-i}{z^4} \text{ is analytic except at } z=0.$$

$$\text{ii) } f(z) = \frac{3\pi^2}{z^3 + 4\pi^2 z} \text{ is analytic except at}$$

$$z^3 + 4\pi^2 z = 0 \Rightarrow z(z^2 + 4\pi^2) = 0$$

$$\Rightarrow z = 0 ; z^2 = -4\pi^2$$

$$\Rightarrow z = 0 ; z = \pm 2\pi i$$

$$\text{iii) } f(z) = \frac{1}{z - z^5} \text{ is analytic except at}$$

$$z - z^5 = 0 \Rightarrow z(1 - z^4) = 0$$

$$\Rightarrow z(1 - z^2)(1 + z^2) = 0$$

$$\Rightarrow z = 0, \pm 1, \pm i$$

Q) Show that if  $f(z)$  is analytic in a domain  $D$  and  $|f(z)| = k = \text{constant}$  in  $D$ , then  $f(z) = \text{Constant}$  in  $D$ .  
 (PT an analytic function of constant absolute value is constant)

Proof :  $f(z) = u + iv$  is analytic  
 $\Rightarrow u_x = v_y \quad \& \quad u_y = -v_x \quad \text{--- (1)}$

$$|f(z)| = k \Rightarrow \sqrt{u^2 + v^2} = k$$

$$\Rightarrow u^2 + v^2 = k^2 \quad \text{--- (2)}$$

Differentiating (2) partially w.r.t  $x$ ,

$$2u \cdot u_x + 2v v_x = 0$$

$$u u_x + v v_x = 0 \quad \text{--- (3)}$$

Differentiating (2) partially w.r.t  $y$ ,

$$2u \cdot u_y + 2v v_y = 0$$

$$u u_y + v v_y = 0 \quad \text{--- (4)}$$

Since  $u_x = v_y \quad \& \quad u_y = -v_x$

$$(3) \Rightarrow u u_x - v u_y = 0 \quad \text{--- (5)}$$

$$(4) \Rightarrow u u_y + v u_x = 0 \quad \text{--- (6)}$$

To eliminate  $u_y$ , multiply (5) by  $u$  and (6) by  $v$  and add.

$$\therefore (5) \times u + (6) \times v \Rightarrow (u^2 + v^2) u_x = 0$$

$$\Rightarrow u_x = 0 \quad [\because u^2 + v^2 = k^2 \neq 0]$$

To eliminate  $u_x$ , multiply (5) by  $-v$  and (6) by  $u$  and add.

$$\therefore (5)x - v + (6)yu \Rightarrow (u^2 + v^2)u_y = 0$$

$$\Rightarrow u_y = 0 \quad [\because u^2 + v^2 = k^2 \neq 0]$$

$$\therefore u_x = 0 \text{ and } u_y = 0 \Rightarrow u = c_1 \text{ (constant)}$$

By C-R equations,

$$u_x = 0 \Rightarrow v_y = 0 \quad \& \quad u_y = 0 \Rightarrow v_x = 0$$

$$\therefore v_x = 0 \text{ and } v_y = 0 \Rightarrow v = c_2 \text{ (constant)}$$

$$\therefore f(z) = \underline{u + iv = c_1 + i c_2 = \text{constant.}}$$

Q) If  $f(z) = u + iv$  is analytic, prove that  $u = c_1$  and  $v = c_2$  are families of curves cutting orthogonally.

Proof: Consider the two families of curves

$$u(x, y) = c_1 \quad \text{--- (1)}$$

$$\text{and } v(x, y) = c_2 \quad \text{--- (2)}$$

Differentiating (1) w.r.t  $x$ , we get

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dx} = 0$$

$$\text{or } \frac{dy}{dx} = \frac{-\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}}$$

$\therefore$  Slope of the tangent to the curve  $u = c_1$  is  $\frac{dy}{dx} = m_1(\text{say})$

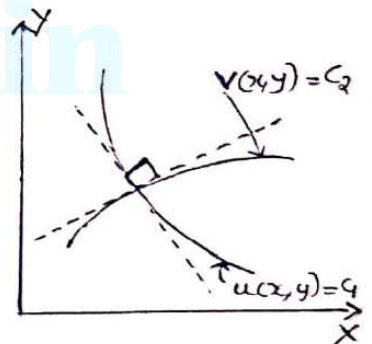
$$\text{i.e., } m_1 = \frac{-\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}}$$

Differentiating (2) w.r.t  $x$ , we get

$$\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-\frac{\partial v}{\partial x}}{\frac{\partial v}{\partial y}} = m_2(\text{say})$$

$\therefore$  Slope of the tangent to the curve  $v = c_2$  is

$$m_2 = \frac{-\frac{\partial v}{\partial x}}{\frac{\partial v}{\partial y}}$$



$$\therefore m_1 \cdot m_2 = \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} \cdot \frac{\frac{\partial v}{\partial x}}{\frac{\partial v}{\partial y}} \quad \text{--- (3)}$$

Since  $f(z)$  is analytic,  $u$  and  $v$  satisfy C-R equations

$$\text{i.e., } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\therefore \text{from (3), } m_1 \cdot m_2 = \frac{\frac{\partial v}{\partial y}}{-\frac{\partial v}{\partial x}} \cdot \frac{\frac{\partial v}{\partial x}}{\frac{\partial v}{\partial y}} = -1$$

Thus the product of the slopes of the curves (1) & (2) is  $-1$ . Hence the curves intersecting at right angles, i.e. they form an orthogonal system.

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## Harmonic Functions:

A real valued function  $\phi(x, y)$  of two variables  $x$  &  $y$  that has continuous second order partial derivatives and satisfy the Laplace's equation

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad (\Leftrightarrow \phi_{xx} + \phi_{yy} = 0)$$

is called a harmonic function.

Theorem : If  $f(z) = u(x, y) + iv(x, y)$  is analytic in a domain  $D$  then both  $u$  and  $v$  satisfy Laplace's equation

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (\Leftrightarrow u_{xx} + u_{yy} = 0)$$

$$\nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad (\Leftrightarrow v_{xx} + v_{yy} = 0)$$

in  $D$  and have continuous second partial derivatives in  $D$ .

(i.e., if  $f(z) = u + iv$  is analytic, then both  $u$  &  $v$  are harmonic)

Proof :  $f(z) = u + iv$  is analytic. Then by C-R eqs

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{--- (1)} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{--- (2)}$$

Differentiating (1) partially w.r.t  $x$  & (2) w.r.t  $y$ ,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{--- (3)}$$

$$\frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x} \quad \text{--- (4)}$$

The derivative of an analytic function is itself analytic.

$\therefore u$  and  $v$  have continuous partial derivatives of all orders; in particular, the mixed second derivatives are equal. i.e.,  $\frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x}$

$$\therefore (3) + (4) \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore u$  is harmonic.

Similarly, differentiating (1) w.r.t  $y$  & (2) w.r.t  $x$ , we get

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2} \quad \text{--- (5)}$$

$$\frac{\partial^2 u}{\partial x \partial y} = -\frac{\partial^2 v}{\partial x^2} \quad \text{--- (6)}$$

$$(5) - (6) \Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad \left[ \because \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \right]$$

$\therefore$   $v$  is harmonic.

Note: If two harmonic functions ' $u$ ' and ' $v$ ' satisfy the C-R equations in a domain  $D$ , they are the real and imaginary part of an analytic function  $f$  in  $D$ .

(ie  $u$  and  $v$  are harmonic &  $u_x = v_y$ ;  $u_y = -v_x$   
 $\Rightarrow f = u + iv$  is analytic)

Then ' $v$ ' is said to be the harmonic conjugate of ' $u$ ' in  $D$ .

Problems: (To find a harmonic conjugate function by the C-R equations)

1. Verify that  $u = x^2 - y^2 - y$  is harmonic and find a harmonic conjugate function  $v$  of  $u$ .

Ans:  $u = x^2 - y^2 - y$

$$\therefore u_x = 2x, \quad u_y = -2y - 1$$

$$u_{xx} = 2, \quad u_{yy} = -2$$

$$\therefore \nabla^2 u = u_{xx} + u_{yy} = 2 - 2 = 0 \Rightarrow u \text{ is harmonic}$$

By C-R equations,

$$v_y = u_x = 2x \quad \text{--- (1) and}$$

$$v_x = -u_y = 2y + 1 \quad \text{--- (2)}$$

Integrating  $v_y$  w.r.t  $y$ , we get (keeping  $x$  constant)

$$v = \int v_y dy = 2xy + h(x) \quad \text{--- (3)}$$

Differentiating this  $v$  w.r.to  $x$ , (partially)

$$\left. \begin{aligned} v_x &= 2y + h'(x) \\ (2) \Rightarrow v_x &= 2y + 1 \end{aligned} \right\} \Rightarrow \begin{aligned} 2y + h'(x) &= 2y + 1 \\ \Rightarrow h'(x) &= 1 \end{aligned}$$

Integrating,  $h(x) = x + c$

Substituting in (3), we get

$v = 2xy + x + c$ , harmonic conjugate of  $u$

Note: The corresponding analytic function is  $f(z) = u + iv$

$$\begin{aligned} f(z) &= x^2 - y^2 - y + i(2xy + x + c) \\ &= x^2 + i2xy - y^2 + i(x + iy) + ic \\ &= (x + iy)^2 + i(x + iy) + ic \\ &= \underline{\underline{z^2 + iz + ic}} \end{aligned}$$

2 Check whether  $v = (2x-1)y$  is harmonic. Find the harmonic conjugate and find the corresponding analytic function  $f(z) = u(x, y) + iv(x, y)$

Ans:  $v = (2x-1)y$

$$v_x = \frac{\partial v}{\partial x} = 2y \quad ; \quad v_{xx} = \frac{\partial^2 v}{\partial x^2} = 0$$

$$v_y = \frac{\partial v}{\partial y} = 2x-1 \quad ; \quad v_{yy} = \frac{\partial^2 v}{\partial y^2} = 0$$

$$\therefore \nabla^2 v = v_{xx} + v_{yy} = 0 + 0 = 0 \Rightarrow v \text{ is harmonic.}$$

By C-R equations,

$$u_x = v_y = 2x-1 \quad \text{--- (1) and}$$

$$u_y = -v_x = -2y \quad \text{--- (2)}$$

Integrating  $u_x$  w.r.t  $x$ , we get (keeping  $y$  constant)

$$u = \int (2x-1) dx = \frac{2x^2}{2} - x + h(y) \quad \text{--- (3)}$$

Differentiating this 'u' w.r.t y partially,

$$\left. \begin{aligned} u_y &= h'(y) \\ (2) \Rightarrow u_y &= -2y \end{aligned} \right\} \Rightarrow \begin{aligned} h'(y) &= -2y \\ h(y) &= \int -2y dy = -y^2 + C \end{aligned}$$

substituting  $h(y)$  in (3), we get

$$u = x^2 - x - y^2 + C$$

The corresponding analytic function is

$$\begin{aligned} f(z) &= u + iv \\ &= x^2 - x - y^2 + C + i(2x - 1)y \\ &= x^2 - y^2 + i2xy - x - iy + C \\ &= (x + iy)^2 - (x + iy) + C \\ &= \underline{\underline{z^2 - z + C}} \end{aligned}$$

3. Determine the constants 'a' and 'b' so that the function is harmonic and find its harmonic conjugate.

i)  $u = ax^3 + bxy$       ii)  $u = e^{-Tx} \cos ay$

Ans : i)  $u = ax^3 + bxy$

$$u_x = 3ax^2 + by \quad ; \quad u_{xx} = 6ax$$

$$u_y = bx \quad ; \quad u_{yy} = 0$$

Since  $u$  is harmonic

$$u_{xx} + u_{yy} = 0 \Rightarrow 6ax = 0$$

$$\Rightarrow a = 0, \text{ b can be any real no.}$$

$$\therefore u = bxy \Rightarrow u_x = by, \quad u_y = bx$$

By C-R equations,

$$\therefore v_y = u_x = by$$

Integrating  $v_y$  w.r.t y, we get-

$$v = \frac{by^2}{2} + h(x) \quad \text{--- (1)}$$

$$\Rightarrow v_x = h'(x) \quad \left. \begin{array}{l} \text{By C-R equations, } v_x = -u_y = -bx \\ \Rightarrow h'(x) = -bx \\ \therefore h(x) = -\frac{bx^2}{2} + c \end{array} \right\}$$

$$\therefore v = \underline{\underline{\frac{by^2}{2} + -\frac{bx^2}{2} + c}} \quad \text{from (1)}$$

$$\text{(ii) } u = e^{-\pi x} \cos ay$$

$$u_x = -\pi e^{-\pi x} \cos ay \quad ; \quad u_{xx} = \pi^2 e^{-\pi x} \cos ay$$

$$u_y = -e^{-\pi x} \sin ay \cdot a \quad ; \quad u_{yy} = -a^2 e^{-\pi x} \cos ay$$

$$u_{xx} + u_{yy} = 0 \Rightarrow (\pi^2 - a^2) e^{-\pi x} \cos ay = 0$$

$$\Rightarrow \pi^2 - a^2 = 0$$

$$\Rightarrow a = \pi$$

$$\therefore u = e^{-\pi x} \cos \pi y$$

$$u_x = -\pi e^{-\pi x} \cos \pi y \quad ; \quad u_y = -\pi e^{-\pi x} \sin \pi y$$

By C-R equations,

$$v_y = u_x = -\pi e^{-\pi x} \cos \pi y$$

$$\left. \begin{array}{l} \text{Integrating} \\ \text{w.r.t } y \end{array} \right\} v = -\pi e^{-\pi x} \frac{\sin \pi y}{\pi} + h(x) = \underline{\underline{-e^{-\pi x} \sin \pi y + h(x)}} \quad (1)$$

Differentiating  $v$  w.r.t  $x$ ,

$$v_x = \pi e^{-\pi x} \sin \pi y + h'(x)$$

$$v_x = -u_y \Rightarrow \pi e^{-\pi x} \sin \pi y + h'(x) = \pi e^{-\pi x} \sin \pi y$$

$$\Rightarrow h'(x) = 0$$

$$\left. \begin{array}{l} \text{Integrating} \\ \text{w.r.t } x \end{array} \right\} \Rightarrow h(x) = c$$

Substituting  $h(x)$  in (1)

$$\underline{\underline{v = -e^{-\pi x} \sin \pi y + c}}$$

## Milne-Thompson Method

This method can be alternatively applied to find the analytic function  $f(z)$  when only real part or imaginary part is given.

Let  $f(z) = u + iv$  be the analytic function

$$\text{Then } f'(z) = u_x + iv_x$$

Step I: If  $u$  is given, take  $f'(z) = u_x - iv_y$

If  $v$  is given, take  $f'(z) = v_y + iv_x$

Step II: Replace ' $x$ ' by  $z$  and ' $y$ ' by 0 in  $f'(z)$

Step III: Integrate  $f'(z)$  w.r.t  $z$ , we get  $f(z)$

Once  $f(z)$  is known, we can determine the conjugate functions  $u$  &  $v$  in terms of  $x$  &  $y$ .

1. Find the analytic function whose imaginary part is
- $$v(x, y) = \log(x^2 + y^2) + x - 2y.$$

Ans: Let  $f(z) = u + iv$  be the analytic function with

$$v = \log(x^2 + y^2) + x - 2y$$

Then  $f'(z) = u_x + iv_x = v_y + iv_x$  [ $\because v$  is given]

$$= \left( \frac{2y}{x^2 + y^2} - 2 \right) + i \left( \frac{2x}{x^2 + y^2} + 1 \right)$$

Using Milne-Thompson method, replace  $x$  by  $z$  and  $y$  by 0, we obtain

$$f'(z) = (0 - 2) + i \left( \frac{2z}{z^2} + 1 \right)$$

$$= i \frac{2}{z} + (i - 2)$$

$$v_y = \frac{2y}{x^2 + y^2} - 2$$

$$v_x = \frac{2x}{x^2 + y^2} + 1$$

Integrating w.r.t  $z$ , we get

$$\underline{f(z) = 2i \ln z + (i-2)z + C}$$

$$\left[ \int \frac{1}{z} dz = \ln z \right]$$

2. Prove that  $u = \sin x \cosh y$  is harmonic. Hence find its harmonic conjugate.

Ans :  $u = \sin x \cosh y$

$$u_x = \cos x \cosh y ; u_{xx} = -\sin x \cosh y$$

$$u_y = \sin x \sinh y ; u_{yy} = \sin x \cosh y$$

$$\therefore u_{xx} + u_{yy} = 0 \Rightarrow u \text{ is harmonic.}$$

Let  $f(z) = u + iv$  be the analytic function with

$$u = \sin x \cosh y$$

$$\begin{aligned} \text{Then } f'(z) &= u_x + iv_x \quad [\because u \text{ is given}] \\ &= u_x - iu_y \end{aligned}$$

$$= \cos x \cosh y - i \sin x \sinh y$$

By Milne-Thompson method, replace  $x$  by  $z$  and  $y$  by  $0$

$$f'(z) = \cos z \cdot \cosh 0 - i \sin z \sinh 0$$

$$= \cos z$$

Integrating w.r.t  $z$ , we get

$$f(z) = \sin z + C$$

$$= \sin(x+iy) + C$$

$$= \sin x \cos(iy) + \cos x \sin(iy) + C$$

$$= \sin x \cosh y + i \cos x \sinh y + C$$

$$= u + iv$$

$$\Rightarrow \underline{u = \sin x \cosh y \text{ \& } v = \cos x \sinh y + C}$$

Results

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\cosh x) = \sinh x$$

$$\cosh 0 = \frac{e^0 + e^{-0}}{2} = 1$$

$$\sinh 0 = \frac{e^0 - e^{-0}}{2} = 0$$

3. Find the real part of an analytic function whose imaginary part is  $v = e^{-x}(x \cos y + y \sin y)$ . Also find the corresponding analytic function.

Ans:  $f'(z) = v_y + i v_x$

$$v_x = \frac{\partial v}{\partial x} = e^{-x}(\cos y) + (x \cos y + y \sin y)(-e^{-x})$$

$$v_y = \frac{\partial v}{\partial y} = e^{-x}(-x \sin y + y \cos y + \sin y)$$

$$\begin{aligned} \therefore f'(z) \bigg|_{(x=z, y=0)} &= e^{-z}(0) + i [e^{-z} + z e^{-z}(-1)] \\ &= i e^{-z}(1-z) \end{aligned}$$

Integrating w.r. to  $z$ , we get

$$f(z) = i \left[ (1-z) \frac{e^{-z}}{-1} - (-1) e^{-z} \right] + C$$

$$= i \left[ (z-1) e^{-z} + e^{-z} \right] + C$$

$$= i \left[ z e^{-z} \right] + C$$

To find the real & imaginary part of  $f(z)$ , put  $z = x + iy$

$$\therefore f(z) = i \left[ (x + iy) e^{-(x + iy)} \right] + C$$

$$= i \left[ (x + iy) e^{-x} \cdot e^{-iy} \right] + C \quad [e^{i\theta} = \cos \theta + i \sin \theta]$$

$$= i \left[ (x + iy) e^{-x} (\cos y - i \sin y) \right] + C$$

$$= i e^{-x} (x \cos y - ix \sin y + iy \cos y + y \sin y) + C$$

$$= e^{-x} \left[ (x \sin y - y \cos y) + i (x \cos y + y \sin y) \right] + C$$

$$= e^{-x} (x \sin y - y \cos y) + i e^{-x} (x \cos y + y \sin y)$$

$$= u + i v$$

$$\Rightarrow u = \underline{e^{-x} (x \sin y - y \cos y) + C}$$

### Home work

1. Are the following functions harmonic? If yes, find a corresponding analytic function  $f(z) = u(x, y) + i v(x, y)$ .

i)  $u = \frac{-x}{x^2 + y^2}$

ii)  $v = e^{-x} \sin 2y$

iii)  $v = 3x^2y - y^3$

2. Find the harmonic conjugate of  $u(x, y) = x^3 - 3xy^2 - 5y$ .

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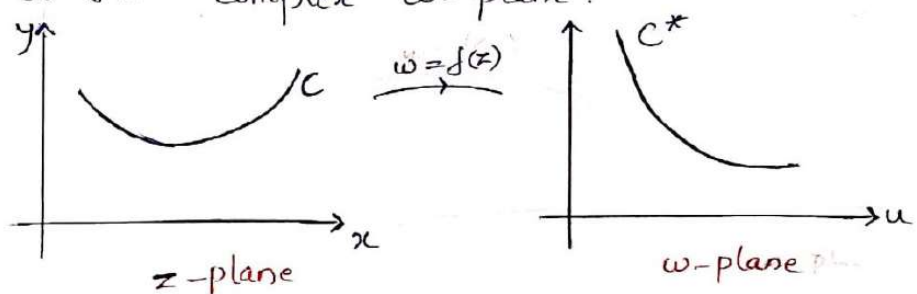
# CONFORMAL MAPPING (Geometry of Analytic Functions)

A real function  $y=f(x)$  can be represented graphically by a curve in the  $xy$ -plane

Since a complex function  $w=f(z)$  (ie,  $u+iv=f(x+iy)$ ) involves four real variables  $x, y$ , (independent variables)  $u(x, y)$  &  $v(x, y)$  (dependent variables), a four dimensional region is required to represent it graphically. As it is not possible, we choose two complex planes and call them  $z$ -plane and  $w$ -plane.

In the  $z$ -plane, we plot the point  $z=x+iy$  and in the  $w$ -plane, plot the corresponding point  $w=u+iv$  (known as the image of ' $z$ ' with respect to ' $f$ '). More generally, for the points of a curve  $C$  in the  $z$ -plane the image points form the image of  $C$  in the  $w$ -plane.

Thus the function  $w=f(z)$  defines a mapping or transformation of its domain of definition  $D$  in the complex  $z$ -plane onto its range of values in the complex  $w$ -plane.



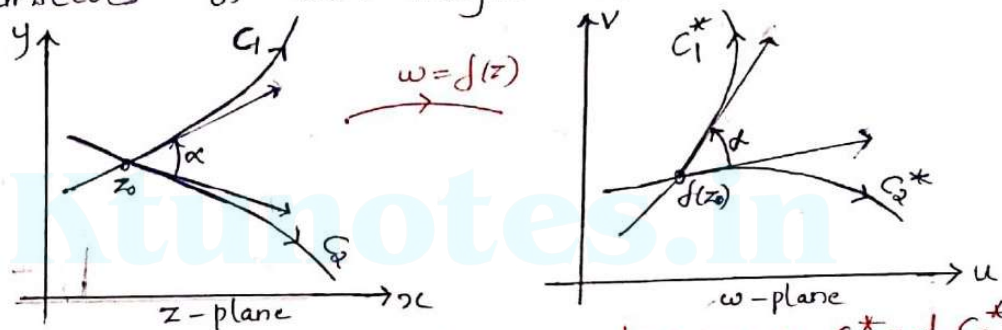
Critical point : of a function  $w=f(z)$  is a point  $z_0$  where  $f'(z_0) = 0$

## Conformal Mapping

A mapping  $w = f(z)$  is called conformal if it preserves angles between oriented curves in magnitude as well as in sense.

The angle  $\alpha$  ( $0 \leq \alpha \leq \pi$ ) between two intersecting curves  $C_1$  and  $C_2$  is defined to be the angle between their oriented tangents at the intersection point  $z_0$ .

Conformality means that the images  $C_1^*$  and  $C_2^*$  of  $C_1$  and  $C_2$  make the same angle as the curves themselves in both magnitude and direction.



Curves  $C_1$  and  $C_2$  and their respective images  $C_1^*$  and  $C_2^*$  under a conformal mapping  $w = f(z)$

### Theorem (Condition for conformality)

A mapping  $w = f(z)$  is conformal at each point  $z_0$  where  $f(z)$  is analytic and  $f'(z_0) \neq 0$ .

i.e., analytic function  $f(z)$  is conformal except at its critical points.

Examples: i) Consider  $f(z) = \cos z \Rightarrow f'(z) = -\sin z$   
 $f'(z) = 0 \Rightarrow z = n\pi$ ,  $n \in \mathbb{Z}$  are the critical points of  $\cos z$

ii)  $f(z) = e^z$  has no critical point

iii)  $f(z) = z^n$ ;  $n > 1$ ;  $z = 0$  is a critical point

iv)  $f(z) = z$  has no critical point.

## Some Important Conformal Mappings

### I Mapping $w = f(z) = z^2$

$w = z^2$  is conformal everywhere except at  $z=0$

[ $\because f(z) = z^2$  is analytic everywhere &  $f'(z) = 0 \Rightarrow z=0$   
(critical point)]

#### Using polar co-ordinates

Let  $z = re^{i\theta}$  and  $w = Re^{i\phi}$

Then  $w = z^2 \Rightarrow Re^{i\phi} = (re^{i\theta})^2 = r^2 e^{i2\theta}$

$$\Rightarrow \boxed{R = r^2 \text{ and } \phi = 2\theta}$$

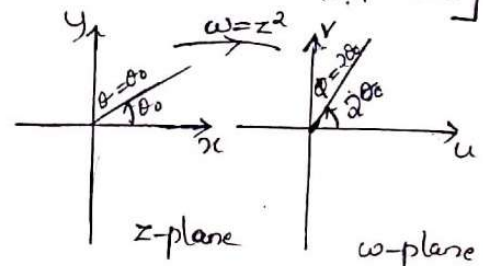
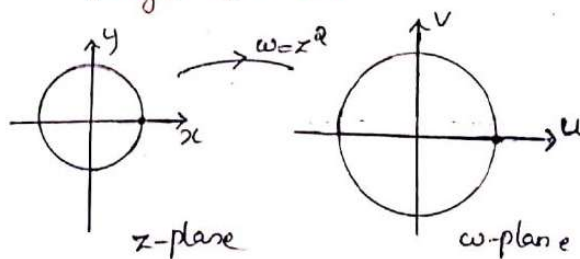
i.e., modulus of  $w$  is  $r^2$  and argument of  $w$  is  $2\theta$

Hence by the mapping  $w = z^2$

- Circles having the polar form  $r = r_0$  (i.e.,  $x^2 + y^2 = r_0^2$ ) in the  $z$ -plane are mapped onto circles  $R = r_0^2$

$$[r = r_0 \Rightarrow r^2 = r_0^2 \Rightarrow R = r_0^2]$$

- Rays  $\theta = \theta_0$  in the  $z$ -plane are mapped onto rays  $\phi = 2\theta_0$  in the  $w$ -plane [ $\because \theta = \theta_0 \Rightarrow 2\theta = 2\theta_0 \Rightarrow \phi = 2\theta_0$ ]



- The upper half of the  $z$ -plane ( $0 \leq \theta \leq \pi$ ) maps into the entire  $w$ -plane  $0 \leq \phi \leq 2\pi$
- The lower half plane ( $-\pi \leq \theta \leq 0$ ) maps into the entire  $w$ -plane.

## Problems

1. Sketch or graph the region and its image under the mapping  $w = z^2$

i)  $|z| \leq \frac{1}{2}$  ,  $-\frac{\pi}{8} < \text{Arg } z < \frac{\pi}{8}$

Ans: Let  $z = re^{i\alpha}$  &  $w = Re^{i\phi}$

$$\therefore w = z^2 \Rightarrow Re^{i\phi} = r^2 e^{i2\alpha}$$

$$\Rightarrow R = r^2 , \phi = 2\alpha$$

$$\therefore |z| \leq \frac{1}{2} \Rightarrow r \leq \frac{1}{2}$$

$$\Rightarrow r^2 \leq \frac{1}{4} \quad (\text{squaring})$$

$$\Rightarrow R \leq \frac{1}{4}$$

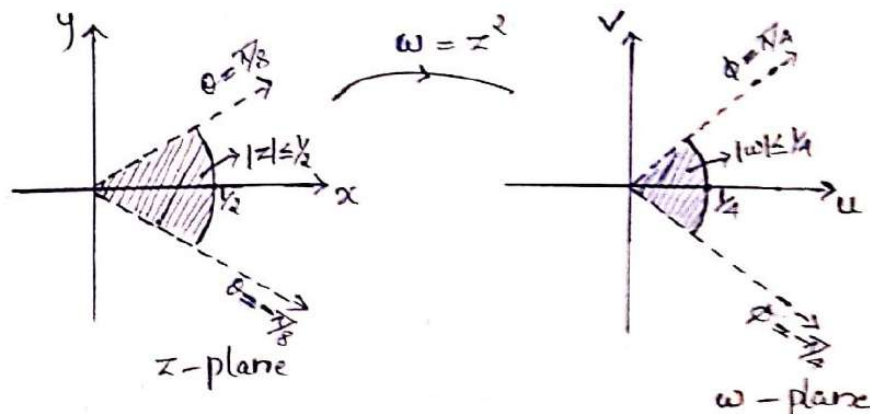
$$\Rightarrow |w| \leq \frac{1}{4} \rightarrow \text{(Closed circular disk centre - 0 \& radius - } \frac{1}{4} \text{)}$$

$$-\frac{\pi}{8} < \text{Arg } z < \frac{\pi}{8} \Rightarrow -\frac{\pi}{8} < \theta < \frac{\pi}{8}$$

$$\Rightarrow -\frac{\pi}{4} < 2\theta < \frac{\pi}{4} \quad (\text{Multiplying by 2})$$

$$\Rightarrow -\frac{\pi}{4} < \phi < \frac{\pi}{4} \rightarrow \text{(region between rays } \phi = -\frac{\pi}{4} \text{ \& } \phi = \frac{\pi}{4} \text{)}$$

$$\Rightarrow -\frac{\pi}{4} < \text{Arg } w < \frac{\pi}{4}$$



$$ii) \quad 2 < |z| < 3, \quad |\arg z| < \pi/4$$

Ans: Let  $z = re^{i\theta}$  &  $w = Re^{i\phi}$

$$\therefore w = z^2 \Rightarrow Re^{i\phi} = r^2 e^{i2\theta}$$

$$\Rightarrow R = r^2, \quad \phi = 2\theta$$

$$2 < |z| < 3 \Rightarrow 2 < r < 3$$

$$\Rightarrow 2^2 < r^2 < 3^2 \quad (\text{squaring})$$

$$\Rightarrow 4 < R < 9$$

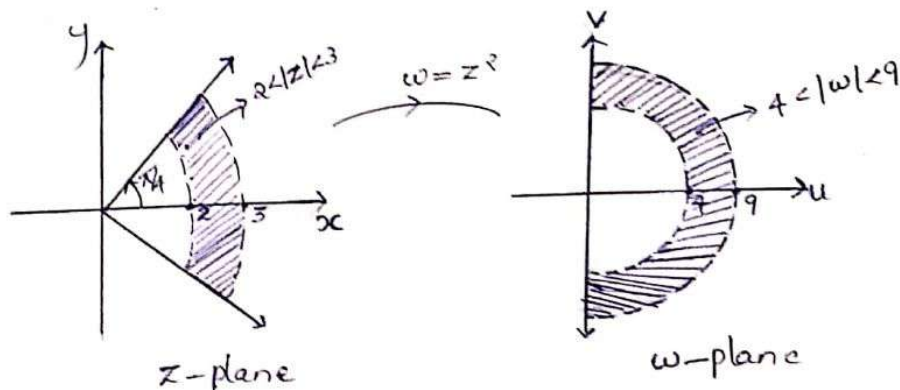
$$\Rightarrow 4 < |w| < 9 \rightarrow (\text{open annulus centred at } 0 \text{ \& radii } 2 \text{ \& } 3)$$

$$|\arg z| < \pi/4 \Rightarrow -\pi/4 < \theta < \pi/4$$

$$\Rightarrow -\pi/2 < 2\theta < \pi/2 \quad (\text{Multiplying by } 2)$$

$$\Rightarrow -\pi/2 < \phi < \pi/2$$

$$\Rightarrow -\pi/2 < \arg w < \pi/2 \rightarrow (\text{region between rays } \phi = \pi/2 \text{ \& } \phi = -\pi/2)$$



$$iii) \quad 1 \leq |z| \leq 3/2; \quad \pi/6 \leq \theta \leq \pi/3$$

Ans: Let  $z = re^{i\theta}$  &  $w = Re^{i\phi}$

$$\therefore w = z^2 \Rightarrow Re^{i\phi} = r^2 e^{i2\theta}$$

$$\Rightarrow R = r^2, \quad \phi = 2\theta$$

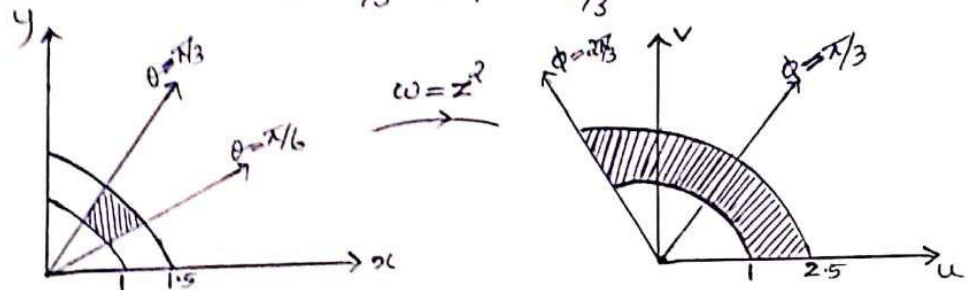
$$1 \leq |z| \leq \frac{3}{2} \Rightarrow 1 \leq r \leq \frac{3}{2}$$

$$\Rightarrow 1 \leq r^2 \leq \frac{9}{4} \quad (\text{squaring})$$

$$\Rightarrow 1 \leq R \leq \frac{9}{4}$$

$$\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3} \Rightarrow \frac{\pi}{3} \leq 2\theta \leq \frac{2\pi}{3}$$

$$\Rightarrow \frac{\pi}{3} \leq \phi \leq \frac{2\pi}{3}$$



In Cartesian co-ordinates, let

$$z = x+iy \quad \text{and} \quad w = u+iv$$

Then  $w = z^2 \Rightarrow u+iv = (x+iy)^2 = x^2 - y^2 + i2xy$

$$\Rightarrow \boxed{u = x^2 - y^2, \quad v = 2xy}$$

Problems :

1. Find the images of
- i)  $x=0$  (imaginary axis or y-axis)
  - ii)  $y=0$  (real axis or x-axis)
  - iii)  $x=c$  (st. line parallel to y-axis)
  - iv)  $y=k$  (st. line parallel to x-axis)

in the  $z$ -plane under the transformation  $w = z^2$

Ans: Let  $z = x+iy$  &  $w = u+iv$

$$\therefore w = z^2 \Rightarrow u+iv = (x+iy)^2 = x^2 - y^2 + i2xy$$

$$\Rightarrow u = x^2 - y^2; \quad v = 2xy$$

$$i) \quad \left. \begin{aligned} x=0 &\Rightarrow u = x^2 - y^2 = -y^2 \\ v = 2xy &= 0 \end{aligned} \right\} \Rightarrow (u, v) = (-y^2, 0)$$

points on the negative  $u$ -axis of  $w$ -plane

ie, the  $y$ -axis is mapped onto negative  $u$ -axis  
(imaginary axis)  $\downarrow$  (-ve real axis)

$$ii) \quad \left. \begin{aligned} y=0 &\Rightarrow u = x^2 - y^2 = x^2 \\ v = 2xy &= 0 \end{aligned} \right\} \Rightarrow (u, v) = (x^2, 0)$$

points on the positive  $u$ -axis of  $w$ -plane

ie, the  $x$ -axis is mapped onto positive  $u$ -axis  
(real axis)  $\downarrow$  (+ve real axis)

$$iii) \quad \begin{aligned} x=c &\Rightarrow u = x^2 - y^2 = c^2 - y^2 \quad \text{--- (1)} \\ &\& \quad v = 2xy = 2cy \quad \text{--- (2)} \end{aligned}$$

Eliminating  $y$

$$\text{From (2), } y = \frac{v}{2c}$$

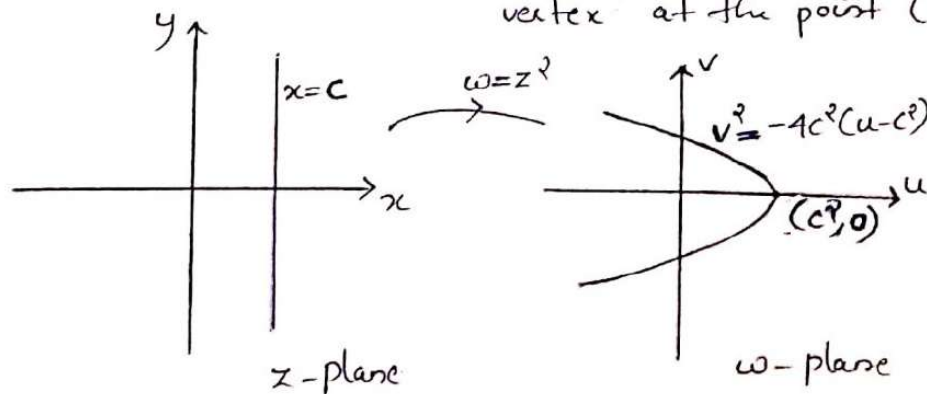
$$\Rightarrow u = c^2 - \frac{v^2}{4c^2} \quad \text{from (1)}$$

$$\Rightarrow \frac{v^2}{4c^2} = c^2 - u$$

$$\Rightarrow v^2 = 4c^2 (c^2 - u)$$

$$\Rightarrow v^2 = -4c^2 (u - c^2)$$

$\hookrightarrow$  parabola opens to the left with vertex at the point  $(c^2, 0)$



$$\text{iv) } y=k \Rightarrow u = x^2 - y^2 = x^2 - k^2 \quad (3)$$

$$v = 2xy = 2xk \quad (4)$$

Eliminating  $x$ ,

$$(4) \Rightarrow x = \frac{v}{2k}$$

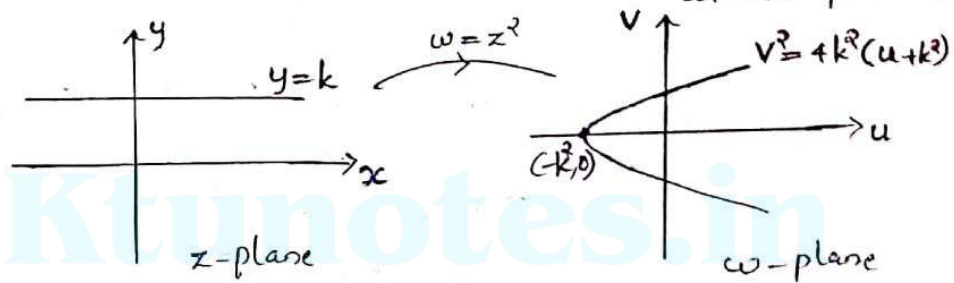
$$\text{Sub. in (3), } u = \frac{v^2}{4k^2} - k^2$$

$$\Rightarrow \frac{v^2}{4k^2} = u + k^2$$

$$\Rightarrow v^2 = 4k^2(u + k^2)$$

$$\Rightarrow v^2 = 4k^2(u - (-k^2))$$

↳ parabola opens to the right with vertex at the point  $(-k^2, 0)$



3. Determine the region of the  $w$ -plane into which the triangular region bounded by  $x=1$ ,  $y=1$  and  $x+y=1$  is mapped by  $w=z^2$

Ans: Take  $z = x+iy$  &  $w = u+iv$

$$\therefore w = z^2 \Rightarrow u+iv = (x+iy)^2 = x^2 - y^2 + i2xy$$

$$\Rightarrow u = x^2 - y^2 \text{ and } v = 2xy$$

$$\text{i) } x=1 \text{ maps into } u = 1 - y^2 ; v = 2y$$

Eliminating  $y$ ,

$$v = 2y \Rightarrow y = \frac{v}{2}$$

$$\therefore u = 1 - \frac{v^2}{4} \Rightarrow \frac{v^2}{4} = 1 - u$$

$$\Rightarrow v^2 = -4(u-1)$$

which is a left handed parabola with vertex at  $(1, 0)$

ii)  $y=1$  maps into  $u = x^2 - 1, v = 2x$

Eliminating  $x$ ,

$$v = 2x \Rightarrow x = \frac{v}{2}$$

$$\therefore u = \frac{v^2}{4} - 1 \Rightarrow \frac{v^2}{4} = u + 1$$

$$\Rightarrow v^2 = 4(u+1) = 4(u-(-1))$$

which is a right handed parabola with vertex at  $(-1, 0)$

iii)  $x+y=1 \Rightarrow y=1-x$

$$\therefore u = x^2 - y^2 = x^2 - (1-x)^2 = -1 + 2x \quad \text{--- (1)}$$

$$v = 2xy = 2x(1-x) = 2x - 2x^2$$

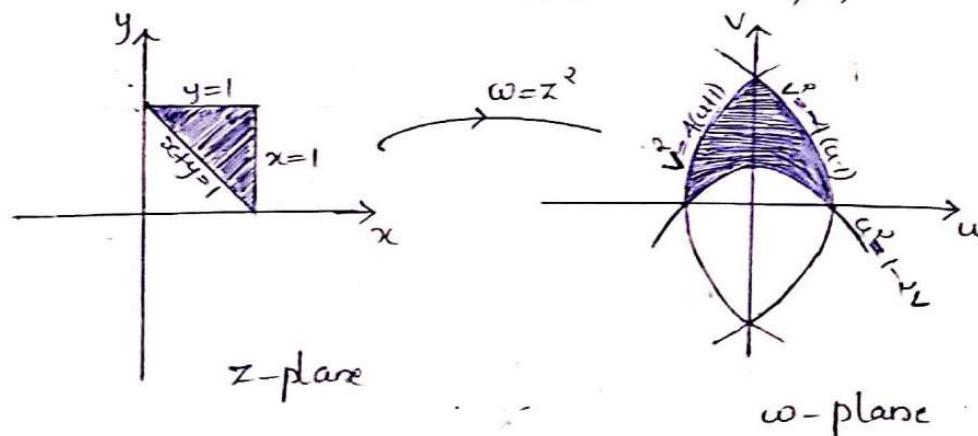
$$u, v = 2 \left[ \frac{u+1}{2} - \left( \frac{u+1}{4} \right)^2 \right]$$

$$= \frac{2}{4} [2(u+1) - (u+1)^2]$$

$$v = \frac{1}{2} (1 - u^2)$$

$$\Rightarrow 2v = 1 - u^2$$

$\Rightarrow u^2 = 1 - 2v = -2(v - \frac{1}{2})$ , a parabola whose vertex is at  $(0, \frac{1}{2})$  opening downwards



## II Mapping $w = f(z) = e^z$

$w = e^z$  is conformal everywhere since  $e^z$  is analytic everywhere and has no critical point  
 $[ \because f'(z) = e^z \neq 0 \text{ for any } z ]$

Put  $z = x + iy$  and  $w = Re^{i\phi}$

$$\text{Then } w = e^z \Rightarrow Re^{i\phi} = e^{x+iy} = e^x \cdot e^{iy}$$

$$\Rightarrow \boxed{R = e^x \text{ and } \phi = y}$$

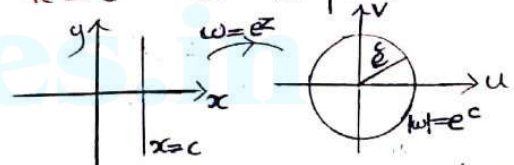
i.e., modulus of  $w$  is  $e^x$  and argument of  $w$  is  $y$

Under  $w = e^z$

1. Vertical straight line  $x=c$  (constant) in  $z$ -plane maps onto the circle  $R=e^c$  in  $w$ -plane

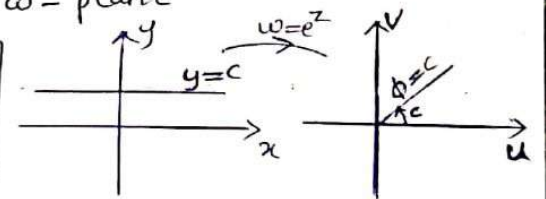
$$\begin{aligned} [\because x=c &\Rightarrow e^x = e^c \\ &\Rightarrow R = e^c \\ &\Rightarrow |w| = e^c \end{aligned}$$

(circle with centre at the origin & radius  $e^c$ )



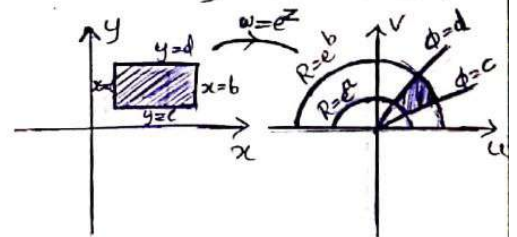
2. Horizontal straight line  $y=c$  in the  $z$ -plane maps onto the ray  $\phi=c$  in the  $w$ -plane

$$\begin{aligned} [ w = e^z &\Rightarrow R = e^x \text{ \& } \phi = y \\ \therefore y=c &\Rightarrow \phi = c \end{aligned}$$

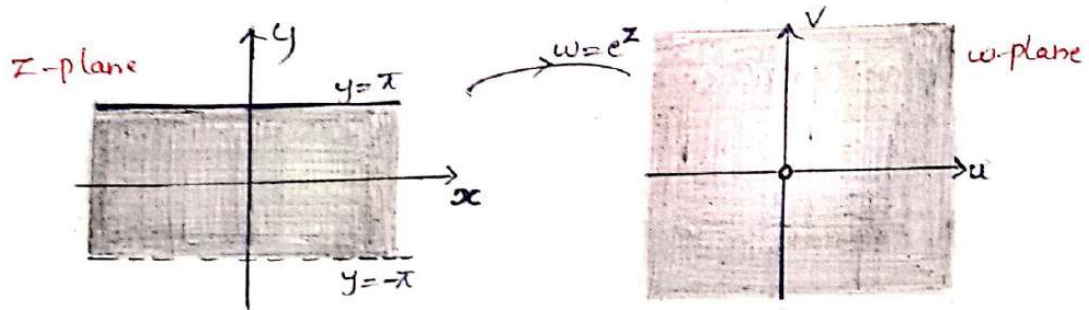


3. The rectangular region given by  $a \leq x \leq b$ ,  $c \leq y \leq d$  is mapped onto a region bounded by circles and rays as shown

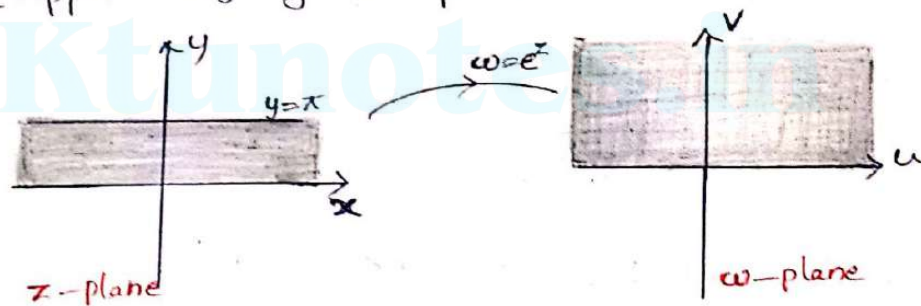
$$\begin{aligned} [\because a \leq x \leq b &\Rightarrow e^a \leq e^x \leq e^b \\ &\Rightarrow e^a \leq R \leq e^b \\ c \leq y \leq d &\Rightarrow e^a \leq |w| \leq e^b \\ &\Rightarrow c \leq \phi \leq d \end{aligned}$$



4.  $-\pi < y \leq \pi$  (or  $-\pi < \text{Arg } z \leq \pi$  - Fundamental region of  $e^z$ ) in the  $z$ -plane is mapped onto the entire  $w$ -plane without the origin  $w=0$   
 [The origin  $w=0$  is not the image of any point  $z$  because  $e^z = 0$  for any  $z$ ]



5. The region  $0 < y \leq \pi$  (upper half of the fundamental region) in the  $z$ -plane is mapped onto  $0 < \phi \leq \pi$  (upper half of  $w$ -plane)



6. The left half plane  $x < 0$  is mapped to inside the unit disk  $|w| < 1$  and the right half plane  $x > 0$  to outside the unit disk  $|w| > 1$

$$\begin{cases} x < 0 \Rightarrow e^x < e^0 \Rightarrow R < 1 \\ x > 0 \Rightarrow e^x > e^0 \Rightarrow R > 1 \end{cases}$$

### Home work

1. Find the image of 1<sup>st</sup> quadrant under the mapping  $w = z^2$ .
2. Find the image of the triangular region bounded by  $x=1$ ,  $y=0$  and  $y=x$  under the transformation  $w = z^2$ .
3. Find and graph the images of the regions under the mapping  $w = e^z$ 
  - i)  $-1 \leq x \leq 3$  ,  $-\pi \leq y \leq \pi$
  - ii)  $-1 < x < 1$  ,  $-\pi/2 < y < \pi/2$

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III

## Linear fractional Transformation $w = \frac{1}{z}$

In polar forms, let

$$z = r e^{i\theta} \quad \text{and} \quad w = R e^{i\phi}$$

$$\text{Then } w = \frac{1}{z} \Rightarrow R e^{i\phi} = \frac{1}{r e^{i\theta}}$$

$$\Rightarrow R e^{i\phi} = \frac{1}{r} e^{-i\theta}$$

$$\Rightarrow R = \frac{1}{r} ; \phi = -\theta$$

Thus for a general  $z$ , the image  $w = \frac{1}{z}$  can be found geometrically by marking  $|w| = R = \frac{1}{r}$  on the segment from  $0$  to  $z$  and then reflecting the mark in the real axis.

### Results

1. Unit circle  $|z|=1$  is mapped on to the unit circle  $|w|=1$   
 $|z|=r=1 \Rightarrow R = \frac{1}{r} = \frac{1}{1} = 1 \Rightarrow |w|=1$
2. Show that  $w = \frac{1}{z}$  maps every straight line or circle onto a circle or straight line.

Proof:

Every st. line or circle in the  $z$ -plane can be written as

$$A(x^2 + y^2) + Bx + Cy + D = 0 \quad (1) \quad (A, B, C, D \text{ are real})$$

$A = 0$  gives a straight line and

$A \neq 0$  gives a circle.

In terms of  $z$  and  $\bar{z}$

$$A z \bar{z} + B \left( \frac{z + \bar{z}}{2} \right) + C \left( \frac{z - \bar{z}}{2i} \right) + D = 0 \quad (2)$$

$$\begin{aligned} z &= x + iy \\ \bar{z} &= x - iy \\ \Rightarrow x &= \frac{z + \bar{z}}{2} \\ y &= \frac{z - \bar{z}}{2i} \end{aligned}$$

$$w = \frac{1}{z} \Rightarrow z = \frac{1}{w} \quad \& \quad \bar{z} = \frac{1}{\bar{w}}$$

$$(2) \Rightarrow A \frac{1}{w\bar{w}} + B \left( \frac{\frac{1}{w} + \frac{1}{\bar{w}}}{2} \right) + C \left( \frac{\frac{1}{w} - \frac{1}{\bar{w}}}{2i} \right) + D = 0$$

Multiplying by  $w\bar{w}$ ,

$$A + B \left( \frac{\bar{w} + w}{2} \right) + C \left( \frac{\bar{w} - w}{2i} \right) + D w\bar{w} = 0$$

In terms of  $u$  and  $v$ ,

$$A + Bu - Cv + D(u^2 + v^2) = 0$$

$$\begin{aligned} w &= u + iv \\ \bar{w} &= u - iv \\ \Rightarrow w\bar{w} &= u^2 + v^2 \\ u &= \frac{w + \bar{w}}{2} \\ v &= \frac{w - \bar{w}}{2i} \end{aligned}$$

This represents a straight line if  $D=0$   
or a circle if  $D \neq 0$  in the  $w$ -plane

### In Cartesian co-ordinates

Let  $w = u + iv$  and  $z = x + iy$

$$\text{Then } w = \frac{1}{z} \Rightarrow z = \frac{1}{w} = \frac{1}{u + iv} = \frac{u - iv}{u^2 + v^2}$$

$$\Rightarrow x + iy = \frac{u - iv}{u^2 + v^2}$$

$$\Rightarrow x = \frac{u}{u^2 + v^2} ; y = \frac{-v}{u^2 + v^2}$$

### Problems

1. Show that under the transformation  $w = \frac{1}{z}$ , the circle  $x^2 + y^2 - 6x = 0$  is transformed into a straight line in the  $w$ -plane.

Ans: Let  $w = \frac{1}{z} \Rightarrow z = \frac{1}{w}$

$$\Rightarrow x = \frac{u}{u^2 + v^2} , y = \frac{-v}{u^2 + v^2}$$

$$\text{Now, } x^2 + y^2 - 6x = 0$$

$$\Rightarrow \frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} - 6 \frac{u}{u^2+v^2} = 0$$

$$\Rightarrow \frac{u^2+v^2}{(u^2+v^2)^2} - \frac{6u}{u^2+v^2} = 0$$

$$\Rightarrow \frac{1}{u^2+v^2} - \frac{6u}{u^2+v^2} = 0$$

$$\Rightarrow \frac{1-6u}{u^2+v^2} = 0$$

$$\Rightarrow 1-6u = 0$$

$$\Rightarrow u = \frac{1}{6}, \text{ a st. line in the } w\text{-plane.}$$

2. Find the image of  $|z-2i|=2$  under the transformation  $w = \frac{1}{z}$

Ans: Let  $w = \frac{1}{z} \Rightarrow z = \frac{1}{w}$

$$\Rightarrow x = \frac{u}{u^2+v^2}, y = \frac{-v}{u^2+v^2}$$

$$\text{Now } |z-2i|=2 \Rightarrow |x+iy-2i|=2$$

$$\Rightarrow |x+i(y-2)|=2$$

$$\Rightarrow \sqrt{x^2+(y-2)^2}=2$$

$$\Rightarrow x^2+(y-2)^2=4$$

$$\Rightarrow x^2+y^2-4y=0$$

$$\Rightarrow \frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} + 4 \frac{v}{u^2+v^2} = 0$$

$$\Rightarrow \frac{u^2 + v^2}{(u^2 + v^2)^2} + \frac{4v}{u^2 + v^2} = 0$$

$$\Rightarrow \frac{1}{u^2 + v^2} + \frac{4v}{u^2 + v^2} = 0$$

$$\Rightarrow 1 + 4v = 0$$

$$\Rightarrow v = -\frac{1}{4}, \text{ st. line in the } w\text{-plane.}$$

3. Find the image of the quadrant  $x > 1, y > 0$  under the transformation  $w = \frac{1}{z}$

Ans: Let  $w = \frac{1}{z} \Rightarrow z = \frac{1}{w}$

$$\Rightarrow x = \frac{u}{u^2 + v^2}, y = \frac{-v}{u^2 + v^2}$$

$$\therefore x > 1 \Rightarrow \frac{u}{u^2 + v^2} > 1$$

$$\Rightarrow u^2 + v^2 - u < 0 \quad \text{--- (1)}$$

[Note:  $u^2 + v^2 + 2gu + 2fv + c = 0$  represents a circle with centre  $(-g, -f)$  and radius  $\sqrt{g^2 + f^2 - c}$ ]

(1)  $\Rightarrow$  interior part of a circle with centre  $(\frac{1}{2}, 0)$  and radius  $\frac{1}{2}$

Now  $y > 0 \Rightarrow \frac{-v}{u^2 + v^2} > 0$

$$\Rightarrow -v > 0$$

$$\Rightarrow v < 0, \text{ half plane below the real axis}$$

$\therefore$  the quadrant  $x > 1, y > 0$  is mapped onto the interior part of the circle passing through origin with centre at  $(\frac{1}{2}, 0)$  and below the real axis.

4. Find the image of the infinite strip  $\frac{1}{4} \leq y \leq \frac{1}{2}$  under the transformation  $w = \frac{1}{z}$ .

Ans: Let  $w = \frac{1}{z} \Rightarrow z = \frac{1}{w}$

$$\Rightarrow x = \frac{u}{u^2+v^2} \text{ and } y = \frac{-v}{u^2+v^2}$$

Now  $\frac{1}{4} \leq y \Rightarrow \frac{1}{4} \leq \frac{-v}{u^2+v^2}$

$$\Rightarrow u^2+v^2 \leq -4v$$

$$\Rightarrow u^2+v^2+4v \leq 0$$

which is the interior part of a circle with centre at  $(0, -2)$  and radius 2.

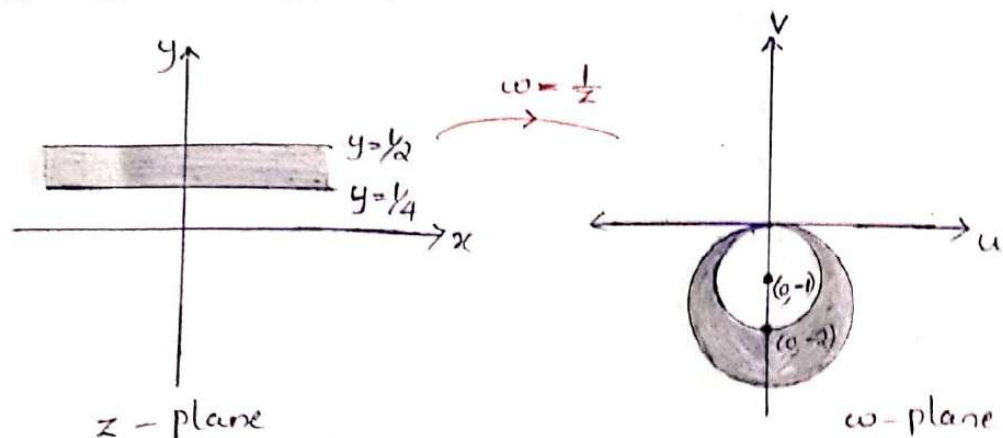
Also  $y \leq \frac{1}{2} \Rightarrow \frac{-v}{u^2+v^2} \leq \frac{1}{2}$

$$\Rightarrow -2v \leq u^2+v^2$$

$$\Rightarrow u^2+v^2+2v \geq 0$$

which is the exterior part of a circle (passing through origin) with centre at  $(0, -1)$  & radius 1.

Thus the infinite strip  $\frac{1}{4} \leq y \leq \frac{1}{2}$  is mapped onto the region between the circles  $u^2+v^2+4v=0$  and  $u^2+v^2+2v=0$ .



5. Find the image of  $x=k$  under  $w = \frac{1}{z}$

Ans:  $w = \frac{1}{z} \Rightarrow z = \frac{1}{w} = \frac{1}{u+iv}$   
 $\Rightarrow x+iy = \frac{u-iv}{u^2+v^2}$   
 $\Rightarrow x = \frac{u}{u^2+v^2}$

Now  $x=k \Rightarrow \frac{u}{u^2+v^2} = k$   
 $\Rightarrow u^2+v^2 = \frac{u}{k}$   
 $\Rightarrow u^2 - \frac{u}{k} + v^2 = 0$   
 $\Rightarrow u^2 - \frac{u}{k} + \frac{1}{4k^2} - \frac{1}{4k^2} + v^2 = 0$   
 $\Rightarrow \left(u - \frac{1}{2k}\right)^2 + v^2 = \frac{1}{4k^2}$

Which is a circle with centre  $\left(\frac{1}{2k}, 0\right)$  & radius  $\frac{1}{2k}$

### Fixed Points

Fixed points of a mapping  $w = f(z)$  are points that are mapped onto themselves i.e., they are kept fixed under the mapping.

i.e., fixed points are given by  $w = f(z) = z$

eg: i)  $w = z$  (identity mapping) has every point as a fixed point.

ii)  $w = \frac{1}{z}$  has two fixed points  $\left[\because \frac{1}{z} = z \Rightarrow z^2 - 1 = 0\right]$   
 $\Rightarrow z = \pm 1$

iii)  $w = az$  has one fixed point  $[az = z \Rightarrow z(a-1) = 0]$   
 $\Rightarrow z = 0$

iv)  $w = z+b$  (where  $b$  is real or complex constant) has no fixed points

IV

## Mapping $w = \sin z$

$$\begin{aligned} u+iv &= w = \sin z = \sin(x+iy) \\ &= \sin x \cos iy + \cos x \sin iy \\ &= \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

$$\Rightarrow u = \sin x \cosh y \quad ; \quad v = \cos x \sinh y$$

Note: Since  $\sin z$  is periodic with period  $2\pi$ , the mapping is not one-to-one if we consider it in the full  $z$ -plane. We restrict  $z$  to the vertical strip  $-\pi/2 \leq x \leq \pi/2$

Note:  $w = \sin z$  not conformal where  $f'(z) = \cos z = 0$   
 $\Rightarrow z = \pm \pi/2$

## Results

1. Image of vertical line  $x = \pm c$

$$w = \sin z \Rightarrow u = \sin x \cosh y \quad ; \quad v = \cos x \sinh y$$

$$\therefore x = \pm c \Rightarrow u = \sin c \cosh y \quad ; \quad v = \cos c \sinh y$$

$$\Rightarrow \frac{u}{\sin c} = \cosh y, \quad \frac{v}{\cos c} = \sinh y$$

$$\Rightarrow \frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c} = \cosh^2 y - \sinh^2 y = 1$$

Thus  $x = \pm c$  are mapped to hyperbolas

$$\frac{u^2}{\sin^2 c} - \frac{v^2}{\cos^2 c} = 1 \quad \text{with foci } (\pm 1, 0)$$

Note: For the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ , foci  $(\pm c, 0)$  where  $c = \sqrt{a^2 + b^2}$

For the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , foci  $(\pm c, 0)$  where  $c = \sqrt{a^2 - b^2}$

## 2. Image of horizontal lines $y = \pm c$

$$w = \sin z \Rightarrow u = \sin x \cosh y ; v = \cos x \sinh y$$

$$\therefore y = \pm c \Rightarrow u = \sin x \cosh c ; v = \cos x \sinh c$$

$$\Rightarrow \frac{u}{\cosh c} = \sin x ; \frac{v}{\sinh c} = \cos x$$

$$\Rightarrow \frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c} = \sin^2 x + \cos^2 x = 1$$

Thus  $y = \pm c$  are mapped to ellipses

$$\frac{u^2}{\cosh^2 c} + \frac{v^2}{\sinh^2 c} = 1 \text{ with foci } (\pm 1, 0)$$

## 3. Exceptional cases: $x = -\pi/2$ & $x = \pi/2$

$$x = \pi/2 \Rightarrow u = \sin \pi/2 \cosh y , v = \cos \pi/2 \sinh y$$

$$\Rightarrow u = \cosh y , v = 0$$

$$\Rightarrow u \geq 1 , v = 0$$

$$\text{Also } x = -\pi/2 \Rightarrow u = \sin(-\pi/2) \cosh y ; v = \cos(-\pi/2) \sinh y$$

$$\Rightarrow u = -\sinh y ; v = 0$$

$$\Rightarrow u = -\cosh y , v = 0$$

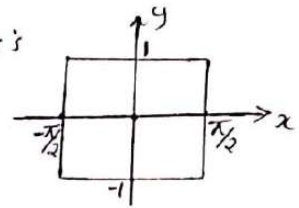
$$\Rightarrow u \leq -1 , v = 0$$

$\therefore$  vertical lines  $x = -\pi/2$  ,  $x = \pi/2$  are folded onto  $u \leq -1$  and  $u \geq 1$  ( $v=0$ ) respectively.

4. Image of the rectangle  $-\pi/2 \leq x \leq \pi/2, -1 \leq y \leq 1$

Ans: The rectangle in the  $z$ -plane is

$$w = \sin z \Rightarrow \begin{aligned} u &= \sin x \cosh y \\ v &= \cos x \sinh y \end{aligned}$$



Consider the upper side:  $y=1, -\pi/2 \leq x \leq \pi/2$

$$\therefore u = \sin x \cosh 1, \quad v = \cos x \sinh 1$$

$$\text{Here } v = \cos x \sinh 1 \geq 0 \quad \left[ \begin{aligned} &\because \cos x \geq 0 \text{ for } -\pi/2 \leq x \leq \pi/2 \\ &\sinh 1 = \frac{e - 1/e}{2} \geq 0 \end{aligned} \right]$$

$\therefore$  the upper side of rectangle is mapped onto the (upper) semi-ellipse  $\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$

Similarly, the lower side:  $y=-1, -\pi/2 \leq x \leq \pi/2$

is mapped onto the (lower) semi-ellipse  $\frac{u^2}{\cosh^2 1} + \frac{v^2}{\sinh^2 1} = 1$

Consider vertical side:  $x=\pi/2, -1 \leq y \leq 1$

$$u = \sin \pi/2 \cosh y = \cosh y; \quad v = \cos \pi/2 \sinh y = 0$$

$$\begin{aligned} \text{Since } -1 \leq y \leq 1 &\Rightarrow u \leq \cosh 1 \\ &\Rightarrow 1 \leq u \leq \cosh 1 \quad \& \quad v=0 \end{aligned}$$

||| by vertical side:  $x=-\pi/2, -1 \leq y \leq 1$

$$u = \sin(-\pi/2) \cosh y = -\cosh y, \quad v = \cos(-\pi/2) \sinh y = 0$$

$$\text{Since } -1 \leq y \leq 1 \Rightarrow -\cosh 1 \leq u \leq -1 \quad \& \quad v=0$$

$\therefore$  vertical sides are mapped onto  $-\cosh 1 \leq u \leq -1$  and  $1 \leq u \leq \cosh 1$  ( $v=0$ ) respectively

